Hilbert–Vector–Resolvent Theory

Positive Moments and Two Conditional Routes
to the Riemann Hypothesis

Preprint · Expanded draft 0.1
28 October 2026

Abstract

We develop an introductory framework connecting scalar coefficients, distinguished vectors in Hilbert space, positive moment measures, and analytic resolvents. For a positive contraction T and a unit vector v, the moments $\langle v, T^n v\rangle$ have nonnegative Bernstein weights; conversely, normalized Bernstein positivity produces a probability measure on $[0,1]$ and a concrete multiplication-operator model. The associated scaled resolvent is holomorphic on a slit plane and has nonpositive imaginary part in the upper half-plane. We give the finite algebra, moment-limit construction, and analytic continuation argument underlying this correspondence.

The arithmetic application concerns the actual normalized moments of the centered Riemann ξ-function. Their universal zero-cancellation condition, denoted $H0$0, yields two conditional routes to the literal Riemann hypothesis: one through spectral growth of Bernstein variation, and one through a positive measure and its resolvent. The existing Lean development proves both implications and the corresponding equivalences; the present text supplies a broader mathematical exposition and a source concordance. The concrete Hilbert-space packaging is presented as classical mathematics, separately from the declarations already audited. The hypothesis $H0$0 itself remains unproved. Neither an unconditional proof of the Riemann hypothesis nor a finite accepted certificate is asserted.

Keywords. Hilbert space; positive contraction; Hausdorff moments; Bernstein weights; scalar resolvent; Riemann hypothesis; formal verification.

Scope of this draft. "Hilbert–Vector–Resolvent theory" is the organizing name used here for the project. The elementary operator and moment correspondences belong to classical analysis. The purpose is to state the project's objects and proof dependencies precisely, to explain the two existing conditional constructions, and to isolate the exact remaining identification and admissibility problems. No priority claim for the classical correspondences is intended.

Source baseline. The mathematical source and verification record are available in the project repository. The frozen project anchor is `ZETA_HYPERBOLIC_ANCHOR_5`.

Contents and reading guide

Part	Pages
Abstract, notation, principal claims, and introduction	1–6
1. Hilbert-vector models Coefficients, defects, positivity, cyclicity, examples	7–14
2. Bernstein positivity and the moment measure Finite identities, shifted moments, positive functionals	15–22
3. Scalar resolvents and analytic zero exclusion Slit domain, derivatives, signs, continuation, poles	23–30
4. Actual ξ-moments and the two RH routes Normalization, spectral growth, common converse	31–38
5. Weighted models and the unfinished frontier Kernel models, exact identification, finite certificates	39–44
Appendix A. Verification evidence and exact endpoints	45–46
Appendix B. Finite prefixes and weak convergence	47–48
References	49
Appendix C. Source concordance and reproducibility	50

The mathematical narrative runs from general objects to the actual arithmetic application. Part 1 fixes the meaning of a Hilbert-vector model and explains why one vector sees a cyclic subspace rather than an arbitrary ambient operator. Part 2 works directly with real sequences. It derives positivity from finite identities before passing to an infinite moment measure. Part 3 studies the analytic function determined by that measure, with all sign conventions fixed explicitly.

Part 4 specializes the construction to the actual centered ξ-function. Its retained spectral infrastructure is stated with its role and provenance; the shorter resolvent route is then assembled from the preceding arguments. Part 5 returns to the broader program and states the missing equality needed to connect the weighted model to the archived Widder family. The appendices separate compiler evidence from mathematical interpretation and record useful limits of finite testing.

Readers interested primarily in the conditional RH result can begin with the statement on page 4 and Part 4, consulting Parts 2 and 3 for the second route. Readers interested in a rigorous general introduction can read Parts 1–3 independently of the arithmetic application. The bibliography supplies classical background; the source concordance gives the more granular project-level correspondence.

Each numbered subsection is a self-contained stage of the argument. Proofs use standard Hilbert-space, measure-theoretic, and complex-analytic results where named. When a substantial theorem is imported from the retained formal development, it is identified as such rather than represented as an elementary consequence of the surrounding exposition.

Notation and conventions

Symbol	Meaning
$\mathcal{H}, T, v$	Complex Hilbert space, bounded operator, distinguished vector.
$\langle x, y\rangle$	Inner product conjugate-linear in x, linear in y.
u_n	Real normalized moment; $u_0 = 1$ when positivity is used.
$w_{N,j}$	Bernstein weight of degree N, for $0 \le j \le N$.
V_N, C_N	Row variation and row cancellation, respectively.
σ	Borel probability measure on $[0, 1]$.
κ	Positive scaling parameter; actual trace moment in Part 4.
S_κ	$\kappa \int_{[0,1]} (1 + \kappa q z)^{-1}\, d\sigma(q)$.
$\mathbb{H}$	Upper half-plane $\{z \in \mathbb{C} : \operatorname{Im} z > 0\}$.
ξ, Θ	Entire Riemann ξ and its even centered descent.
R	The germ Θ'/Θ at the origin.
$H00$	Semantic admissibility of the actual zero-budget candidate.

All sequence indices start at zero. The binomial coefficient is the ordinary integer binomial coefficient. Finite sums over an empty set are zero. The grid point at degree zero is set to $q_{0,0} = 0$; for $N > 0$, the grid is $q_{N,j} = j/N$. This convention avoids interpreting $0/0$ when defining Bernstein averages.

For a bounded self-adjoint operator, $0 \le T \le I$ means that $\langle x, Tx\rangle$ and $\langle x, (I - T)x\rangle$ are nonnegative for every x. The term *positive contraction* includes both self-adjointness and these inequalities. A bare contraction need not have real or positive scalar coefficients.

The two scalar resolvent conventions must be kept distinct:

$$F(z) = \langle v, (I - zT)^{-1} v\rangle, \qquad S_\kappa(z) = \kappa \langle v, (I + \kappa z T)^{-1} v\rangle.$$

For a positive model, F has nonnegative imaginary part on $\mathbb{H}$, while S_κ has nonpositive imaginary part there. The inequalities need not be strict; the measure δ_0 gives a constant resolvent. "Anti-Herglotz" will mean the latter nonpositive sign condition, with the convention always displayed when used.

An equality of germs means equality throughout some neighborhood of the stated point. Holomorphy on a domain does not assert that this domain is maximal. In particular, the universal slit used for S_κ is a common domain guaranteed for every probability measure on $[0, 1]$, not the singular set of every particular example.

Principal claims and their logical scope

The analytic framework has a general equivalence independent of the zeta function. It is a classical form of the compact moment problem.

Theorem 1 (Positive-model correspondence)**.** *Let $(u_n)_{n\geq 0}$ be real and $u_0 = 1$. The following are equivalent:*

(i) Every Bernstein weight $w_{N,j}$ defined on page 15 is nonnegative.

(ii) There is a Borel probability measure σ on $[0,1]$ such that $u_n = \int q^n \, d\sigma(q)$ for all n.

(iii) There is a positive contraction T on a complex Hilbert space and a unit vector v with $u_n = \langle v, T^n v\rangle$.

The measure in (ii) is unique. One may choose the cyclic model $\mathcal{H} = L^2(\sigma)$, $T = M_q$, $v = 1$.

The implication (i)⇒(ii) is developed in Part 2; (ii)⇒(iii) is the multiplication construction in Part 1; and (iii)⇒(i) follows from the positive operators $T^j(I-T)^{N-j}$. Uniqueness follows from polynomial density.

Theorem 2 (Retained actual conditional endpoints)**.** *For the actual normalized ξ-moment sequence defined in Part 4, the semantic condition $H0$0 is equivalent to the literal Riemann hypothesis. The retained development provides two forward constructions of $H0$0 ⇒ RH and a common converse.*

Here $H0$0 is a universal assertion about all Bernstein rows. Its finite candidate data are $(0,0)$, but its admissibility is not a finite computation. The first route uses the spectral growth theorem for row variation. The second constructs the measure, identifies the actual logarithmic-derivative germ, and excludes nonreal zeros. Precise endpoint types and their audit record appear in Appendix A.

There are three distinct levels of claim in this manuscript. The finite identities and actual conditional endpoints have source counterparts covered by the existing audit. The full complex Hilbert realization and some general explanatory lemmas are proved here as ordinary mathematics, without a claim of new Lean coverage. Finally, unconditional $H0$0, checker soundness and completeness, and the archived Widder identification remain goals. None is promoted to a theorem by the existence of a model with similarly named coefficients.

This separation is mathematically consequential. A positive measure for an arbitrary sequence gives a resolvent for that sequence. To derive a statement about ξ, the sequence must be the actual Taylor data of its logarithmic derivative. That identification is the link which makes the second route an arithmetic argument.

Introduction: from scalar data to an operator model

A sequence often enters analysis in several apparently different forms: as the derivatives of a germ, as moments of a measure, or as matrix coefficients of an operator orbit. The useful question is not whether one can assign these descriptions formally, but which identities and inequalities make them describe the same object. The framework of this preprint begins with the chain

$$u_n = \int_0^1 q^n \, d\sigma(q) = \langle 1, M_q^n 1 \rangle_{L^2(\sigma)}.$$

The moment measure supplies positivity and compact support. The operator supplies norm estimates and geometric defects. The resolvent packages all moments into a single analytic function. Each description therefore exposes a different part of the same structure.

The finite entry point is the Bernstein basis $\binom{N}{j} q^j (1-q)^{N-j}$. If a normalized sequence acts positively on every member of this basis, its row weights form probability distributions on increasingly fine grids. The finite coarsening identity measures how total variation changes from one row to the next. Vanishing cancellation is precisely the condition that all these signed rows are already positive. The passage to a measure then requires an actual limit argument; row positivity alone is not a license to interchange arbitrary infinite operations.

Once the measure exists, its compact support controls denominators:

$$S_\kappa(z) = \kappa \int_0^1 \frac{d\sigma(q)}{1 + \kappa q z}, \qquad z \notin (-\infty, -1/\kappa].$$

The imaginary part of the integrand has a fixed sign in the upper half-plane. This sign becomes a zero-exclusion mechanism only after the resolvent is identified with the logarithmic derivative of an entire function. At a zero of positive multiplicity, that logarithmic derivative has a pole with a positive integer residue. A local downward approach to the pole contradicts the resolvent sign.

The actual ξ application is an instance of this mechanism with carefully normalized Taylor data. It also admits a different route through exponential growth of Bernstein variation. The coexistence of the two routes is useful: one makes the geometry of the canonical zeros visible, while the other makes the positive measure and analytic sign visible. Their common unresolved premise is the same exact universal admissibility statement.

The introduction therefore aims at a disciplined synthesis. Classical Hilbert and moment theory provide the language; the repository gives specific actual constructions and audited endpoints; the remaining program is expressed as explicit assertions whose proofs can be sought independently.

Representation, identification, and the proof architecture

Three operations organize the discussion. *Representation* constructs an object realizing given data. *Identification* proves that these data equal those of a prescribed analytic function. *Closure* converts the resulting analytic information into a statement about zeros. Confusing any two of these operations can hide a missing hypothesis.

For representation, Parts 1 and 2 connect four descriptions:

$$\begin{aligned} \text{nonnegative Bernstein rows} &\iff \text{positive moment measure on } [0,1], \\ \text{positive moment measure} &\iff \text{cyclic positive-contraction model.} \end{aligned}$$

The second equivalence is understood up to unitary equivalence of cyclic models. It is not a uniqueness assertion for all possible ambient spaces containing an unobserved orthogonal summand. The resulting resolvent has prescribed Taylor coefficients and a half-plane sign, independently of any arithmetic interpretation.

For identification, let $\Theta(0) \neq 0$ and assume that the actual derivatives satisfy

$$\left(\frac{\Theta'}{\Theta}\right)^{(n)}(0) = (-1)^n n! \kappa^{n+1} u_n.$$

The same formula holds for $S_\kappa^{(n)}(0)$. Analyticity makes the two germs equal. Their equality is propagated using the holomorphic defect $\Theta' - S_\kappa \Theta$ on the upper half-plane; dividing by Θ before excluding its zeros would require unnecessary care. For closure, the positive-multiplicity pole of Θ'/Θ is incompatible with the anti-Herglotz sign. Conjugation symmetry then handles the lower half-plane.

The first arithmetic route uses a different intermediate invariant:

$$H00 \Longrightarrow V_N = 1 \ (\forall N) \Longrightarrow \text{subexponential variation} \Longrightarrow \text{RH}.$$

The last implication uses the retained actual spectral extraction theorem. The second route replaces that forward spectral-growth step by the positive-measure construction and analytic closure. It shares normalization, direct positivity lemmas, and the converse from RH.

The classical background is the compact Hausdorff moment problem, the spectral theory of bounded self-adjoint operators, and elementary complex analysis [2, 1, 5]. The arithmetic normalization is compatible with the classical ξ-function [6]. Formal proof infrastructure is supplied by Lean and mathlib [3, 4]; the project's precise theorem scope is fixed by its source and audit records [7].

1.1 Hilbert triples and conventions

The operator language begins with a complex Hilbert space $\mathcal{H}$, a bounded linear operator $T\colon \mathcal{H} \to \mathcal{H}$, and a distinguished vector v. Throughout, the inner product is conjugate-linear in its first argument and linear in its second:

$$\langle ax, by\rangle = \overline{a}\, b\langle x, y\rangle, \qquad \|x\|^2 = \langle x, x\rangle.$$

This convention fixes the signs of the resolvent identities below. The zero vector gives an identically zero scalar function; normalization will usually mean $\|v\| = 1$, but the elementary estimates retain the factor $\|v\|^2$ so that weighted vectors are included.

Definition 3. A *contractive Hilbert-vector model* is a triple $(\mathcal{H}, T, v)$ with $\|T\| \le 1$. It is *positive* when T is self-adjoint and $0 \le T \le I$. Its scalar orbit coefficients are

$$c_n = \langle v, T^n v\rangle, \qquad n \ge 0.$$

Positivity in this definition is operator positivity, not positivity of individual matrix entries in a chosen basis. In particular, it is invariant under unitary change of coordinates. The construction also singles out a vector: the operator alone does not specify the scalar coefficients.

The natural observable subspace is

$$\mathcal{H}_v = \overline{\operatorname{span}}\{v, Tv, T^2v, \ldots\}.$$

It is invariant under T by boundedness. If $T = T^*$, then it is reducing. Indeed, if $x \perp \mathcal{H}_v$ and $y \in \mathcal{H}_v$, then $\langle y, Tx\rangle = \langle Ty, x\rangle = 0$, since $Ty \in \mathcal{H}_v$. Thus $\mathcal{H}_v^{\perp}$ is invariant as well. The vector is called *cyclic* if $\mathcal{H}_v = \mathcal{H}$.

For self-adjoint T, every scalar moment is therefore already a moment of the restriction to $\mathcal{H}_v$:

$$\langle v, T^n v\rangle = \langle v, (T|_{\mathcal{H}_v})^n v\rangle.$$

The complement is invisible to these data. This observation will later prevent an unwarranted conclusion about the entire spectrum of an operator reconstructed from one scalar function. These Hilbert-space definitions are classical mathematical background; they are stronger than the minimal Gram-orbit interface used in the audited source.

1.2 Coefficient bounds and the two defects

Let $(\mathcal{H}, T, v)$ be a contractive model. Iteration preserves the norm bound, so $\|T^n v\| \le \|v\|$ for every n. Combining this fact with Cauchy–Schwarz immediately bounds the scalar orbit:

$$|c_n| = |\langle v, T^n v\rangle| \le \|v\|\,\|T^n v\| \le \|v\|^2.$$

It is useful to retain the two reasons for this bound separately. Write $x = T^n v$, and define the contraction and Gram defects by

$$A = \|v\|^2(\|v\|^2 - \|x\|^2), \qquad B = \|v\|^2\|x\|^2 - |\langle v, x\rangle|^2.$$

Contractivity gives $A \ge 0$, whereas Cauchy–Schwarz gives $B \ge 0$. The following identity is purely algebraic.

Proposition 4. *For every $n \ge 0$,*

$$\|v\|^4 - |c_n|^2 = A + B.$$

In particular, the coefficient defect is a sum of two nonnegative quantities. Equivalently, it is the Euclidean norm-square of $(\sqrt{A}, \sqrt{B}) \in \mathbb{R}^2$.

Proof. Add the displayed definitions of A and B. The two occurrences of $\|v\|^2\|x\|^2$ cancel. Nonnegativity justifies the square roots and the norm-square description. □

This decomposition separates two possible losses. The orbit may lose norm under iteration, producing a positive contraction defect. Even when its norm is preserved, the orbit vector may cease to be aligned with the distinguished vector, producing a positive Gram defect. No self-adjointness is needed for either statement.

If $v \ne 0$, equality $|c_n| = \|v\|^2$ occurs precisely when $T^n v = \lambda v$ for some scalar λ with $|\lambda| = 1$. To see this, equality in the coefficient bound forces both defects to vanish. Equality in Cauchy–Schwarz makes x a scalar multiple of v, and equality of norms determines the modulus of that scalar. The converse follows by substitution.

The source development isolates the real counterpart of this argument using a norm-square function and a squared Cauchy inequality [7]. Such an interface is sufficient for the coefficient estimate. It does not, by itself, construct a Hilbert space or prove linearity of its orbit map.

1.3 Neumann series and disk analyticity

For $|z| < 1$, the operator series

$$R_z = \sum_{n=0}^{\infty} z^n T^n$$

converges in operator norm, because $\|z^n T^n\| \le |z|^n$. Multiplying its finite partial sums by $I - zT$ gives

$$(I - zT) \sum_{n=0}^{N} z^n T^n = I - z^{N+1} T^{N+1}.$$

The remainder tends to zero in norm. The same calculation on the other side proves $R_z = (I - zT)^{-1}$ and $\|R_z\| \le (1 - |z|)^{-1}$.

Proposition 5. *The scalar generating function of a contractive model is holomorphic on the unit disk and satisfies*

$$F(z) = \sum_{n=0}^{\infty} c_n z^n = \langle v, (I - zT)^{-1} v \rangle, \qquad |F(z)| \le \frac{\|v\|^2}{1 - |z|}.$$

Proof. The map $A \mapsto \langle v, Av \rangle$ is a continuous linear functional on bounded operators, of norm at most $\|v\|^2$. Apply it to the operator-norm convergent Neumann series. On each closed disk $|z| \le r < 1$, the geometric estimates are uniform, so the series is holomorphic there and hence throughout the open disk. □

The argument also controls derivatives. Since T commutes with $I - zT$, it commutes with its inverse. Differentiating $(I - zT)R_z = I$ gives $R_z' = R_z T R_z$. Induction consequently yields

$$F^{(k)}(z) = k!\, \langle v, T^k (I - zT)^{-k-1} v \rangle, \qquad |F^{(k)}(z)| \le \frac{k! \|v\|^2}{(1 - |z|)^{k+1}}.$$

In particular, $F^{(k)}(0) = k! c_k$. The coefficient sequence is therefore the actual Taylor sequence of this analytic function, not merely a formal expression assigned to the model.

The same majorant gives a quantitative approximation by finite data: for $|z| \le r < 1$,

$$\left| F(z) - \sum_{n=0}^{N} c_n z^n \right| \le \frac{\|v\|^2 r^{N+1}}{1 - r}.$$

This estimate is uniform in the model once the vector norm is fixed. It controls truncation inside the disk, without supplying control across its boundary.

The unit disk is a guaranteed domain, not necessarily the maximal one. For example, a strict contraction gives a larger Neumann disk, and spectral information can provide continuation far beyond it. The construction at this stage uses neither a positive measure nor an arithmetic calibration. Those additional structures will be needed to obtain a half-plane sign and an identity with a prescribed function.

1.4 Positive contractions and complete differences

Assume now that $T = T^*$, $0 \le T \le I$, and $\|v\| = 1$. The order inequalities mean $0 \le \langle x, Tx\rangle \le \|x\|^2$ for every x. The spectrum of T lies in $[0, 1]$. The continuous functional calculus for bounded self-adjoint operators preserves real nonnegative functions on that interval [1].

Put $u_n = \langle v, T^n v\rangle$. Positivity of T^n and $I - T^n$ gives $0 \le u_n \le 1$, and $u_0 = 1$. More generally, the commuting operators T and $I - T$ satisfy

$$\sum_{r=0}^{k} (-1)^r \binom{k}{r} u_{n+r} = \langle v, T^n (I - T)^k v\rangle \ge 0.$$

The final inequality follows by applying functional calculus to $q^n(1-q)^k$, which is nonnegative for $0 \le q \le 1$. These finite-difference inequalities contain more information than the bound $|u_n| \le 1$.

Proposition 6. *The Bernstein row associated with the moments of a positive model is a nonnegative row of mass one:*

$$w_{N,j} = \binom{N}{j} \langle v, T^j (I - T)^{N-j} v\rangle \ge 0, \qquad \sum_{j=0}^{N} w_{N,j} = 1.$$

Proof. Each weight is nonnegative because its defining polynomial $\binom{N}{j} q^j (1-q)^{N-j}$ is nonnegative on $[0, 1]$. The binomial identity for the commuting pair $(T, I - T)$ shows that the sum of the operators is I, and normalization then gives total mass one. □

The first two difference orders already have recognizable content: $u_n \ge u_{n+1}$ and $u_n - 2u_{n+1} + u_{n+2} \ge 0$. Thus positive models produce decreasing, discretely convex sequences. Higher orders impose further conditions; neither monotonicity nor convexity alone characterizes a Hausdorff moment sequence.

The moments also give positive Hankel forms. For arbitrary complex scalars $a_0, \dots, a_d$,

$$\sum_{i,j=0}^{d} \overline{a_i} a_j u_{i+j} = \left\| \sum_{j=0}^{d} a_j T^j v \right\|^2 \ge 0.$$

Here self-adjointness is used to identify $\langle T^i v, T^j v\rangle$ with u_{i+j}. The identities explain how vector positivity is visible through finite scalar tests. Their converse requires a representation argument: checking finitely many particular rows does not establish all orders, and a formal interface with positive fields does not prove that those fields can be supplied for the intended sequence.

1.5 The concrete multiplication model

Let σ be a Borel probability measure on $I = [0, 1]$. Consider the complex Hilbert space

$$\mathcal{H} = L^2(I, \sigma; \mathbb{C}), \qquad \langle f, g \rangle = \int_I \overline{f(q)} g(q)\, d\sigma(q).$$

Functions here are equivalence classes modulo equality almost everywhere. Multiplication by the bounded coordinate function preserves that equivalence and defines a bounded linear operator:

$$(Tf)(q) = qf(q), \qquad v(q) = 1.$$

Since $|q| \le 1$, one has $\|Tf\|_2 \le \|f\|_2$. The constant function belongs to the space and has norm one.

Proposition 7. *The triple $(L^2(\sigma), T, 1)$ is a positive Hilbert-vector model. For every $n \ge 0$ it realizes the moments of σ:*

$$\langle 1, T^n 1 \rangle = \int_I q^n\, d\sigma(q).$$

Proof. The coordinate q is real, so $\langle Tf, g \rangle = \langle f, Tg \rangle$ for all f, g. A bounded operator defined on the whole space with this symmetry is self-adjoint. Moreover, $\langle f, Tf \rangle = \int q|f|^2\, d\sigma$ lies between zero and $\|f\|_2^2$. Induction gives $T^n 1 = q^n$, proving the moment formula. □

If $z \notin [1, \infty)$, the continuous function $q \mapsto 1 - zq$ has no zero on the compact interval I. Its reciprocal is therefore bounded. Multiplication by that reciprocal is a bounded two-sided inverse of $I - zT$, giving

$$\langle 1, (I - zT)^{-1} 1 \rangle = \int_I \frac{d\sigma(q)}{1 - zq}.$$

Thus the vector resolvent is exactly an integral transform, including at $z = 0$. No choice of basis or eigenvector expansion is involved, and no density assumption on σ is needed. Atomic, singular, and absolutely continuous measures are treated identically.

On a compact subset of this slit domain, $|1 - zq|$ has a positive lower bound uniform in both z and q. Difference quotients and their limits are consequently dominated by an integrable constant. Differentiation under the integral is valid, proving holomorphy directly throughout that domain.

This model is the concrete Hilbert-space interpretation of the probability measure constructed later from positive Bernstein rows. The audited project proves that measure construction and its scalar integral consequences. The present operator packaging is mathematical exposition; it is not an additional Lean construction already covered by the source audit.

1.6 Polynomial density and uniqueness of the measure

Two standard approximation principles make the multiplication model particularly precise. First, complex polynomials restricted to I are uniformly dense in $C(I, \mathbb{C})$ by Stone–Weierstrass. Second, continuous functions are dense in $L^2(I, \sigma)$ for a finite Borel measure on this compact metric space [5]. The latter follows from regularity: approximate indicators by continuous functions between a compact subset and an open superset, and then approximate simple functions.

Since $\sigma(I) = 1$, uniform approximation implies approximation in the Hilbert norm:

$$\|f - p\|_{L^2(\sigma)} \leq \|f - p\|_\infty \qquad (f \in C(I),\ p \text{ a polynomial}).$$

It follows that the polynomial classes are dense in $L^2(\sigma)$. But $p(T)1 = p(q)$, so

$$\overline{\operatorname{span}}\{1, T1, T^2 1, \ldots\} = L^2(I, \sigma).$$

The distinguished vector in the multiplication model is cyclic, even when the measure has atoms or is concentrated on a proper closed subset of the interval.

Proposition 8. *A Borel probability measure on* $[0, 1]$ *is uniquely determined by all its monomial moments.*

Proof. Suppose σ and τ have the same moments. Linearity gives equal integrals for every polynomial. If $p_k \to f$ uniformly, then

$$\left| \int f \, d\sigma - \int p_k \, d\sigma \right| \leq \|f - p_k\|_\infty,$$

and the same bound holds for τ. The two integrals therefore agree for every continuous f. Uniqueness in the Riesz representation theorem implies $\sigma = \tau$. □

All moments are essential in this statement. A finite collection determines integrals only on a finite-dimensional polynomial space and generally permits many representing measures. For instance, the unit mass at $1/2$ and normalized Lebesgue measure on $[0, 1]$ have the same zeroth and first moments. Their second moments are $1/4$ and $1/3$, respectively. Even exact agreement at those first two orders leaves different resolvents at the next Taylor coefficient. The eventual construction of the actual measure thus has two distinct features: existence follows from all-order positivity, while exact moment identification fixes its scalar resolvent uniquely. Neither fact supplies a terminating decision procedure for the all-order positivity condition.

1.7 Spectral measures and the visible operator

For a bounded self-adjoint positive contraction T, the spectral theorem provides a projection-valued measure E supported in $[0,1]$. A unit vector v turns it into a scalar probability measure:

$$\sigma_v(B) = \langle v, E(B)v\rangle = \|E(B)v\|^2.$$

The spectral calculus then gives, for continuous f,

$$\langle v, f(T)v\rangle = \int_{[0,1]} f(q)\, d\sigma_v(q).$$

In particular, the moments and resolvent are those of σ_v. This recovers the integral description from an abstract positive model; the preceding multiplication construction provides the converse direction.

The cyclic restriction is determined up to unitary equivalence. Indeed, define on polynomials

$$U\colon p \longmapsto p(T)v.$$

The moment identities imply $\|p(T)v\|^2 = \int |p(q)|^2\, d\sigma_v(q)$. Thus U respects the null classes in $L^2(\sigma_v)$ and is isometric on the polynomial subspace. Density extends it to a unitary map from $L^2(\sigma_v)$ onto $\mathcal{H}_v$. The identity $U(qp) = T\,U(p)$ on polynomials extends by boundedness, so U intertwines multiplication by q with $T|_{\mathcal{H}_v}$.

There is no corresponding determination of the unobserved orthogonal complement. For any other positive contraction A, the enlarged model

$$(\mathcal{H}\oplus\mathcal{K},\ T\oplus A,\ v\oplus 0)$$

has exactly the same scalar moments and resolvent as the original one. The spectrum of A may add points not present in the cyclic restriction. Consequently a scalar pole or support calculation must be interpreted relative to the represented vector, not automatically as a statement about an arbitrary ambient operator.

This distinction matters when using operator language in a zero problem. Constructing some self-adjoint operator is not enough: one must specify the distinguished vector, prove the exact moment identities, and identify the associated analytic function. The actual-function calibration in the RH application performs that final identification at the scalar level. It is not replaced by the existence of the abstract spectral representation.

1.8 Elementary models and limits of the assumptions

Explicit examples distinguish contractivity, positivity, and actual-function identification without introducing analytic complications.

Example 9 (A single spectral atom)**.** On $\mathbb{C}$, let $T = qI$, $v = 1$, and $0 \le q \le 1$. Then

$$u_n = q^n, \qquad F(z) = \frac{1}{1-qz}, \qquad w_{N,j} = \binom{N}{j} q^j (1-q)^{N-j}.$$

The corresponding measure is δ_q. At $q = 0$ the resolvent is constant; at $q = 1$ it has a pole at $z = 1$. Thus the guaranteed unit disk need not be the maximal domain.

Example 10 (Two atoms)**.** Let $T = \operatorname{diag}(q_1, q_2)$ on $\mathbb{C}^2$ and $v = (\sqrt{t}, \sqrt{1-t})$, where $q_1, q_2, t \in [0, 1]$. Then

$$u_n = tq_1^n + (1-t)q_2^n, \qquad F(z) = \frac{t}{1-q_1 z} + \frac{1-t}{1-q_2 z}.$$

The scalar measure is $t\delta_{q_1} + (1-t)\delta_{q_2}$. If $t = 1$, the second coordinate is invisible to F, even when $q_2 \neq q_1$. For distinct q_1, q_2 and $0 < t < 1$, however, v and Tv are linearly independent: their coordinate determinant is $\sqrt{t(1-t)}(q_2 - q_1) \neq 0$. Hence v is cyclic in this case. Both atoms are visible, illustrating directly why positive weights and distinct support points enter the cyclic reconstruction.

A self-adjoint contraction need not be positive. For $T = -I$ and $v = 1$, the coefficients are $c_n = (-1)^n$ and $F(z) = (1+z)^{-1}$. The coefficient bound and disk analyticity hold, but $c_1 < 0$, so these are not moments of a positive measure on $[0, 1]$. The positivity assumption cannot be recovered from the uniform coefficient estimate.

Self-adjointness also matters for the reconstruction statement. Let S be the unilateral shift on $\ell^2(\mathbb{N})$ and $v = e_0$. The orbit is orthonormal, giving

$$\langle e_0, S^n e_0 \rangle = \begin{cases} 1, & n = 0, \\ 0, & n > 0. \end{cases} \qquad F(z) = 1 \quad (|z| < 1).$$

The vector is cyclic, yet this is the same scalar generating function as the zero operator on $\mathbb{C}$. The two operators are not unitarily equivalent. Scalar orbit moments of a general contraction do not determine mixed orbit inner products. The positive self-adjoint setting supplies exactly that extra structure; the actual arithmetic application still requires its separate calibration even after positivity is known.

2.1 Bernstein rows and mass

For this chapter let $u = (u_k)_{k\geq 0}$ be an arbitrary real sequence. Normalization and positivity will be imposed explicitly when needed. Associate to u the polynomial moment functional

$$\mathcal{M}_u\left(\sum_{k=0}^{d} c_k x^k\right) = \sum_{k=0}^{d} c_k u_k, \qquad B_{N,j}(x) = \binom{N}{j} x^j (1-x)^{N-j}.$$

This functional is defined by finite algebra; no representing measure, continuity assertion, or growth restriction on u is implicit. Applying it to the Bernstein basis gives the weights

$$w^u_{N,j} = \mathcal{M}_u(B_{N,j}) = \binom{N}{j} \sum_{r=0}^{N-j} (-1)^r \binom{N-j}{r} u_{j+r}, \qquad 0 \leq j \leq N. \tag{1}$$

The superscript will be suppressed when the sequence is fixed. Each row depends on only $u_0, \ldots, u_N$. Thus every particular weight is finite data, even though positivity of all rows is an infinite family of conditions.

With the forward-difference convention $\Delta u_j = u_{j+1} - u_j$, the inner sum is $(-1)^{N-j}\Delta^{N-j} u_j$. Since the binomial factor is strictly positive at every in-range index, positivity of all Bernstein rows is equivalent to nonnegativity of every alternating finite difference. The row formulation retains the binomial weights because they organize both conservation of mass and the compatibility between successive degrees. These two identities will control the normalization of the eventual representing measure.

Proposition 11 (Conservation of row mass)**.** *For every real sequence and every $N \geq 0$,*

$$\sum_{j=0}^{N} w^u_{N,j} = u_0.$$

Proof. The binomial theorem gives the polynomial identity $\sum_j B_{N,j}(x) = (x + (1-x))^N = 1$. Apply the linear functional $\mathcal{M}_u$. Equivalently, expansion of the defining finite sum shows that the coefficient of u_k is

$$\sum_{j=0}^{k} \binom{N}{j}(-1)^{k-j}\binom{N-j}{k-j} = \binom{N}{k}\sum_{j=0}^{k}\binom{k}{j}(-1)^{k-j} = \begin{cases} 1, & k = 0, \\ 0, & 1 \leq k \leq N. \end{cases}$$

Here the product of binomial coefficients counts the same successive choices in two orders. All rearrangements are finite, including when $N = 0$. □

The first rows make the sign convention transparent:

$$(w_{0,0}) = (u_0), \qquad (w_{1,0}, w_{1,1}) = (u_0 - u_1, u_1),$$
$$(w_{2,0}, w_{2,1}, w_{2,2}) = (u_0 - 2u_1 + u_2, 2u_1 - 2u_2, u_2).$$

If a positive measure already represents u, these are integrals of nonnegative polynomials. Our later construction reverses that implication: the signs of these finite expressions will produce the measure. Conservation of mass is the normalization mechanism for that reversal. In particular, when $u_0 = 1$, every row has signed mass one regardless of its individual signs. Its absolute mass can nevertheless exceed one; that excess, rather than the signed mass, detects cancellation.

2.2 Exact coarsening and increasing variation

The Bernstein bases at successive degrees satisfy a compatibility relation. It provides a direct comparison of row variation without assuming that any weights are positive. For $0 \le j \le n$, the two binomial identities

$$(n+1-j)\binom{n+1}{j} = (n+1)\binom{n}{j}, \qquad (j+1)\binom{n+1}{j+1} = (n+1)\binom{n}{j}$$

give an identity between polynomials. Factoring out $\binom{n}{j}x^j(1-x)^{n-j}$ from its right-hand side leaves $(n+1)((1-x)+x)$. Consequently,

$$(n+1)B_{n,j} = (n+1-j)B_{n+1,j} + (j+1)B_{n+1,j+1}. \tag{2}$$

The endpoint cases $j = 0, n$ require no separate convention. Applying $\mathcal{M}_u$ proves the exact weighted relation

$$(n+1)w_{n,j} = (n+1-j)w_{n+1,j} + (j+1)w_{n+1,j+1}. \tag{3}$$

The coefficients become nonnegative after division by $n+1$. They describe how adjacent entries of a finer row contribute to one coarser entry. They need not sum to one within that pair: the relevant conservation law is obtained by summing contributions from the whole finer row.

Proposition 12 (Monotonicity of variation)**.** *Define* $V_N = \sum_{j=0}^{N} |w_{N,j}|$. *Then*

$$|u_0| \le V_N \le V_{N+1} \qquad (N \ge 0), \qquad V_0 = |u_0|.$$

Proof. The first inequality is the triangle inequality applied to row mass. Taking absolute values in (3) and summing gives

$$(n+1)V_n \le \sum_{j=0}^{n}\big((n+1-j)|w_{n+1,j}| + (j+1)|w_{n+1,j+1}|\big) = (n+1)V_{n+1}.$$

For an interior finer-row index k, the two coefficients are $n+1-k$ and k, whose sum is $n+1$. The indices 0 and $n+1$ each occur once, again with coefficient $n+1$. Division by this positive integer proves monotonicity. □

Equation (3) also shows that positivity propagates downward in degree: if every entry of row $n+1$ is nonnegative, then so is every entry of row n. Iteration gives positivity of every earlier row from positivity of one later row. This is a finite implication, with the direction fixed by the coarsening identity. It supplies no positivity information about rows of higher degree than those already controlled.

This proof identifies the only possible source of strict growth: the triangle inequality at an adjacent pair. There is no loss from reindexing or from normalization. The next page records that loss exactly, rather than estimating it. This exactness is useful because a zero bound will force equality at every finite step.

2.3 Triangle defects and the cancellation identity

Fix a row transition $n \to n+1$. For $0 \le j \le n$, set

$$a_{n,j} = (n+1-j)w_{n+1,j}, \qquad b_{n,j} = (j+1)w_{n+1,j+1}, \qquad D_{n,j} = |a_{n,j}| + |b_{n,j}| - |a_{n,j} + b_{n,j}|.$$

The quantity $D_{n,j}$ is a nonnegative real number. For real scalars, it vanishes exactly when $a_{n,j}b_{n,j} \ge 0$, including either scalar being zero. Indeed, if the signs agree, absolute value is additive; if the signs differ, the defect is twice the smaller absolute value. Thus the terminology "cancellation" has a literal algebraic meaning. Define the total defect of the transition by

$$C_n = \sum_{j=0}^{n} D_{n,j} \ge 0.$$

Proposition 13 (Exact variation increment)**.** *Every real sequence satisfies*

$$C_n = (n+1)(V_{n+1} - V_n). \tag{4}$$

Proof. Equation (3) implies $|a_{n,j} + b_{n,j}| = (n+1)|w_{n,j}|$. The coefficient count in the preceding proof gives $\sum_j (|a_{n,j}| + |b_{n,j}|) = (n+1)V_{n+1}$. Subtract the sum of the former identities from the latter identity. Every sum is finite, so this is an equality with no convergence issue. □

Two useful consequences retain the full information of the defect:

$$C_n = 0 \iff V_{n+1} = V_n, \qquad V_N = |u_0| + \sum_{n=0}^{N-1} \frac{C_n}{n+1}.$$

The second identity follows by telescoping, with an empty sum when $N = 0$. It separates the initial absolute mass from the accumulation of local cancellation. In particular, a bound on the weighted sum of defects bounds total variation; an assertion that all defects vanish fixes total variation exactly.

When $u_0 = 1$, another exact expression is

$$V_N - 1 = 2\sum_{j=0}^{N} \max\{-w_{N,j}, 0\}.$$

To prove it, subtract row mass from absolute mass and use $|t| - t = 2\max\{-t, 0\}$ term by term. This makes the distinction between signed mass and negative mass explicit. One zero transition does not assert positivity of every row. Positivity at all degrees will follow from zero defects at all transitions together with the initial positive mass.

For example, the normalized initial moments $u_0 = 1$, $u_1 = u_2 = -1$ give rows $(2, -1)$ and $(2, 0, -1)$. Both have variation three, so $C_1 = 0$ despite their negative entries. The earlier transition has $C_0 = 2$. This example isolates why all transitions, including the transition out of the degree-zero row, occur in the zero-budget hypothesis.

The source definitions `cancellationDefect` and `rowCancellation` encode these same weighted triangle defects [7].

2.4 The zero budget and its equivalent forms

For a normalized real sequence, a cancellation budget is a pair of natural numbers $W = (N_0, b)$. Its semantic admissibility means

$$\mathrm{Adm}_u(N_0, b) \quad \Longleftrightarrow \quad \forall n \geq N_0,\ C_n \leq bV_n.$$

Write $H_{00}(u)$ for $\mathrm{Adm}_u(0,0)$. This notation temporarily makes the sequence explicit. In the actual application, H_{00} without an argument always refers to the fixed normalized logarithmic-derivative moments specified in Part 4.

Theorem 14 (Finite identities behind the zero budget)**.** *If $u_0 = 1$, then the following four assertions are equivalent:*

$$\begin{aligned} H_{00}(u) &\iff \forall n \geq 0,\ C_n = 0 \\ &\iff \forall N \geq 0,\ V_N = 1 \\ &\iff \forall N \geq 0\ \forall j \leq N,\ w_{N,j} \geq 0. \end{aligned} \tag{5}$$

Proof. Admissibility gives $C_n \leq 0$ at every index; nonnegativity of defects turns this inequality into equality. Conversely, zero defects satisfy the budget. By (4), zero defects are equivalent to equal successive variations. Since $V_0 = |u_0| = 1$, induction gives $V_N = 1$ at every degree. Constant variation conversely makes every increment vanish.

For a fixed row, conservation of mass yields

$$V_N - 1 = \sum_{j=0}^{N} (|w_{N,j}| - w_{N,j}).$$

Each summand is nonnegative. A finite sum of such terms is zero exactly when every term is zero, and $|t| - t = 0$ exactly when $t \geq 0$. This proves both remaining directions, row by row. □

The role of normalization is precise. It supplies a positive initial mass and identifies minimal variation with one. If $u_0 = c > 0$, the same argument replaces one by c. For the actual sequence, normalization $u_0 = 1$ is already a retained theorem, so no additional normalization hypothesis is attached to the actual implication.

The pair $(0, 0)$ has a finite description, but its admissibility is the universally quantified assertion displayed above. Checking any fixed collection of rows proves only the corresponding finite collection of inequalities. The theorem therefore gives equivalent forms of the common hypothesis, not its unconditional proof. At degree one, positivity says $0 \leq u_1 \leq 1$; at degree two, it additionally says $0 \leq u_2 \leq u_1$ and $u_2 \geq 2u_1 - 1$. These are useful necessary conditions on the first moments, obtained without analytic machinery. The full condition requires every such finite-difference inequality at every order, so these low-degree constraints do not replace its quantifiers. The actual specializations are present in `RouteOne.lean`, including `H00_iff_all_variation_one` and `H00_iff_actual_weights_nonnegative` [7].

2.5 Finite averages and the shifted-moment recurrence

For a function on the compact interval define the finite Bernstein average using the explicit degree-zero convention

$$q_{N,j} = \frac{j}{N} \quad (N \geq 1), \qquad q_{0,0} = 0, \qquad A_N^u(f) = \sum_{j=0}^{N} w_{N,j}^u f(q_{N,j}).$$

Thus $A_0^u(f) = u_0 f(0)$; a constant monomial always means the constant function one, including its value at zero. No sign hypothesis is imposed in this definition. Let $u_r^+ = u_{r+1}$ be the shifted sequence. Directly from (1),

$$(j+1)w_{N+1,j+1}^u = (N+1)w_{N,j}^{u^+} \qquad (0 \leq j \leq N). \tag{6}$$

Indeed, the inner finite differences on the two sides are identical, and their outer coefficients agree by $(j+1)\binom{N+1}{j+1} = (N+1)\binom{N}{j}$.

For a positive-degree monomial the $j = 0$ contribution at degree $N+1$ vanishes. Reindexing all remaining terms and using (6) to absorb one power of the grid coordinate gives

$$A_{N+1}^u(x^{k+1}) = \sum_{j=0}^{N} w_{N,j}^{u^+} \left(\frac{j+1}{N+1}\right)^k. \tag{7}$$

This formula also holds at $N = 0$. To express its right-hand side through averages at degree N, assume now $N \geq 1$ and use

$$\frac{j+1}{N+1} = \frac{N}{N+1}\frac{j}{N} + \frac{1}{N+1}, \qquad T_{N,k,i} = \binom{k}{i}\left(\frac{N}{N+1}\right)^i \left(\frac{1}{N+1}\right)^{k-i}.$$

The ordinary binomial theorem, followed by interchange of two finite sums, proves the recurrence

$$A_{N+1}^u(x^{k+1}) = \sum_{i=0}^{k} T_{N,k,i} A_N^{u^+}(x^i) \qquad (N \geq 1). \tag{8}$$

Its degree is lower on the right, but its moment sequence is shifted. This matters for the induction on the next page: the inductive statement must quantify over all sequences, not just a single fixed one. The transition coefficients depend only on N, k, i. Although they are nonnegative and sum to one, the averages multiplying them can have either sign. Their convergence will come from induction, without a bound on the variation of the original sequence. There is also a useful exact first-moment check on the reindexing. Putting $k = 0$ in (7) reduces its right-hand side to the mass of a row for u^+, namely u_1. Thus every positive-degree grid recovers the first moment exactly, even when some of its weights are negative. The exceptional grid at degree zero has average zero on x, and does not affect any limiting statement. Stating that exception explicitly avoids silently applying an affine-grid calculation with a zero denominator.

2.6 Monomial convergence without positivity

Theorem 15 (Algebraic moment recovery)**.** *For every real sequence u and every integer $k \geq 0$,*

$$\lim_{N\to\infty} A_N^u(x^k) = u_k. \tag{9}$$

Proof. We use strong induction on k, keeping the sequence universally quantified. At degree zero, row mass gives the exact identity

$$A_N^u(1) = \sum_{j=0}^{N} w_{N,j}^u = u_0 \qquad \text{for every } N \geq 0.$$

Suppose the assertion holds in every degree at most k, for every real sequence. In particular, it holds for the shifted sequence u^+. The elementary scalar limits $N/(N+1) \to 1$ and $1/(N+1) \to 0$ imply, for each $0 \leq i \leq k$,

$$T_{N,k,i} \longrightarrow \begin{cases} 1, & i = k, \\ 0, & i < k, \end{cases} \qquad A_N^{u^+}(x^i) \longrightarrow u_{i+1}.$$

For $i < k$ the exponent $k - i$ is positive; for $i = k$ it is zero and the binomial coefficient is one. Taking limits in the finite sum (8) therefore gives

$$A_{N+1}^u(x^{k+1}) \longrightarrow \sum_{i=0}^{k} \mathbf{1}_{\{i=k\}} u_{i+1} = u_{k+1}.$$

The recurrence is valid for every $N \geq 1$, which is sufficient for this limit. Removing the fixed index shift preserves convergence, and proves the claim at degree $k + 1$. □

The proof does not assume normalization, bounded moments, or nonnegative weights. It does not exchange an infinite series with a limit: for each fixed degree there are finitely many transition terms. It also does not assert convergence for every continuous test function when the weights have unrestricted signs. Such an extension needs uniform control of the functionals, which positivity will provide below.

Linearity immediately extends the conclusion to every polynomial:

$$p(x) = \sum_{k=0}^{d} c_k x^k \quad \Longrightarrow \quad A_N^u(p) \longrightarrow \sum_{k=0}^{d} c_k u_k = \mathcal{M}_u(p).$$

As elementary checks, the recurrence gives $A_N^u(x) = u_1$ for $N \geq 1$ and $A_N^u(x^2) = N^{-1}u_1 + (1 - N^{-1})u_2$ for $N \geq 1$. These are consistent both with degree one and with the limiting second moment.

The absence of growth assumptions is possible because a fixed polynomial degree probes only finitely many initial moments. The recurrence reduces that degree while shifting the sequence, and the induction terminates at row mass. It does not provide estimates uniform in the polynomial degree. Consequently, using this theorem inside an infinite power-series argument requires an additional summability or domination argument; that step is kept separate in the later resolvent analysis.

The general induction is the argument encoded by `bernsteinAverage_monomial_tendsto` in `BernsteinMomentLimit.lean` [7].

2.7 A single ultrafilter and a positive functional

Assume henceforth $u_0 = 1$ and $w^u_{N,j} \geq 0$ for all admissible indices. Each row is now a probability vector. On the real Banach space $C([0,1])$, the finite averages obey

$$A^u_N(\alpha f + \beta g) = \alpha A^u_N(f) + \beta A^u_N(g),$$
$$A^u_N(1) = 1, \qquad f \geq 0 \Longrightarrow A^u_N(f) \geq 0.$$

Moreover, positivity and mass one imply the common norm bound

$$|A^u_N(f)| \leq \sum_{j=0}^{N} w^u_{N,j} |f(q_{N,j})| \leq \|f\|_\infty. \tag{10}$$

Choose one free ultrafilter $\mathcal{U}$ on $\mathbb{N}$ containing every cofinite set, and use this same choice for every test function. For each f, the sequence of averages lies in the compact interval $[-\|f\|_\infty, \|f\|_\infty]$. Compactness gives an ultrafilter limit, and the Hausdorff property of the real line makes it unique. Define

$$L_u(f) = \lim_{N \to \mathcal{U}} A^u_N(f).$$

The use of a single ultrafilter is essential to this construction: it lets addition and scalar multiplication pass to limits in a common limiting process. An arbitrary, separately selected cluster point for each function would not by itself establish linearity.

Proposition 16. *The map $L_u : C([0,1]) \to \mathbb{R}$ is a positive bounded linear functional with $L_u(1) = 1$ and operator norm one.*

Proof. The finite linear identities pass to $\mathcal{U}$-limits because addition and scalar multiplication are continuous. The closed sets $[0, \infty)$ and $[-\|f\|_\infty, \|f\|_\infty]$ preserve, respectively, positivity and the bound (10). The constant sequence $A^u_N(1) = 1$ gives $L_u(1) = 1$. Therefore $\|L_u\| \leq 1$, while evaluation on the unit vector 1 gives the reverse inequality. □

Because $\mathcal{U}$ contains every cofinite set, each ordinary convergent sequence has the same ultrafilter limit. The preceding page therefore yields

$$L_u(x^k) = u_k \qquad (k \geq 0).$$

The source uses `Filter.hyperfilter` for the common choice.

Its `hausdorffPositiveFunctional` is constructed from these limits. This is an existence construction using classical choice; it supplies a coherent functional before any representing measure is introduced. It is useful to distinguish the two compactness inputs. Compactness of $[0,1]$ makes each continuous test function bounded; the resulting compact interval of real values supplies the ultralimit. No subsequence indexed simultaneously for an uncountable set of test functions is selected. The single ultrafilter provides that simultaneous consistency through preservation of finite algebraic operations.

2.8 The Hausdorff measure and uniqueness

Theorem 17 (Probability representation)**.** *If $u_0 = 1$ and every Bernstein weight of u is nonnegative, there is a unique Borel probability measure σ on $[0,1]$ satisfying*

$$u_k = \int_{[0,1]} q^k \, d\sigma(q) \qquad (k \geq 0). \tag{11}$$

Proof. The real Riesz representation theorem on the compact Hausdorff space $[0,1]$ applies to the positive functional L_u. It gives a regular Borel measure with

$$L_u(f) = \int f \, d\sigma \quad (f \in C([0,1])), \qquad \sigma([0,1]) = L_u(1) = 1.$$

Taking monomials and using $L_u(x^k) = u_k$ proves the moment identities. For uniqueness, let σ and τ be two Borel probability measures with these moments. Their integrals agree on polynomials by linearity. Polynomial density in $C([0,1])$ implies that for each continuous f and $\varepsilon > 0$ there is a polynomial p with $\|f - p\|_\infty < \varepsilon$. Hence

$$\left| \int f \, d\sigma - \int f \, d\tau \right| \leq \int |f - p| \, d\sigma + \int |f - p| \, d\tau < 2\varepsilon.$$

The continuous integrals therefore agree. Finite Borel measures on this compact metric space are regular, and uniqueness in Riesz representation gives $\sigma = \tau$. □

The converse is immediate but useful: any probability measure with the displayed moments gives

$$w^u_{N,j} = \int B_{N,j}(q) \, d\sigma(q) \geq 0.$$

Thus normalized Bernstein positivity is exactly the compact-interval moment condition. Uniqueness also removes any dependence of the resulting measure on the free ultrafilter selected in the construction. In fact the entire sequence of averages converges weakly to this measure: polynomial approximation and (10) give $A^u_N(f) \to \int f \, d\sigma$ for every continuous f, by bounding the two approximation errors and then taking the polynomial limit. This also proves directly that the limiting functional is intrinsic to the moments: any bounded linear functional agreeing with it on all monomials agrees on polynomials and then, by continuity, on all continuous functions. The compact support is crucial to this density argument. Moment uniqueness on a noncompact interval would require additional hypotheses and is not being inferred here.

For the actual moments, Theorem 14 supplies positivity from the same hypothesis H_{00}. The source construction `actualHausdorffMeasure` and its moment theorem implement this conditional measure step in `HausdorffMeasure.lean` [7]. The uniqueness and weak convergence arguments just given are mathematical exposition; their inclusion here does not claim a new, separately audited Lean declaration. No unconditional proof of H_{00} has entered the construction.

3.1 The scalar resolvent and its universal domain

Throughout this chapter, σ is a finite positive Borel measure on $I = [0, 1]$, $M = \sigma(I)$, and $\kappa > 0$. Probability normalization is useful for the application, but the analytic arguments require only finiteness. The scalar object attached to these data is

$$S_\kappa(z) = \kappa \int_I \frac{1}{1 + \kappa q z}\, d\sigma(q).$$

The factor κ and the plus sign in the denominator are part of our convention. They determine both the Taylor coefficients and the direction of the imaginary-part inequality. Changing either convention requires changing the later formulas accordingly.

Proposition 18 (Universal denominator domain)**.** *The set on which every denominator is nonzero is exactly*

$$D_\kappa := \{z \in \mathbb{C} :\ 1 + \kappa q z \neq 0 \text{ for all } q \in I\} = \mathbb{C} \setminus (-\infty, -1/\kappa].$$

In particular, it contains the origin and both open half-planes.

Proof. At $q = 0$ the denominator equals one. For $q > 0$, its only zero is

$$z = -\frac{1}{\kappa q} \leq -\frac{1}{\kappa}.$$

Conversely, if $x \leq -1/\kappa$, the number $q = -1/(\kappa x)$ belongs to $(0, 1]$ and makes that denominator zero. These two implications give the claimed equality. For fixed $z \in D_\kappa$, the integrand is continuous on the compact interval, hence bounded and integrable against σ. □

The adjective universal is essential. The displayed slit is determined by all possible parameters in the interval, not by the support of one chosen measure. For the point mass at $a \in (0, 1]$, for example,

$$S_\kappa(z) = \frac{\kappa}{1 + \kappa a z}, \qquad \sigma = \delta_a,$$

which extends meromorphically through every other point of the slit. For $\sigma = \delta_0$ the transform is the constant κ and is entire. Nevertheless, every excluded point is a genuine pole for some point mass. Thus no larger common domain can guarantee a holomorphic transform for all positive measures on I.

This distinction prevents a support bound from being mistaken for an assertion about the exact singularities of the resulting function. We will establish holomorphy on D_κ without assuming the existence, absence, or location of zeros of any entire function to which the transform may later be compared. All integrals here are ordinary complex integrals, equivalently the integrals of the real and imaginary parts. No density with respect to Lebesgue measure is required: discrete, continuous, and mixed measures are treated by the same definition.

3.2 Compact denominator control and continuity

Pointwise nonvanishing alone does not justify an interchange of a limit and an integral. What makes the present construction tractable is the compactness of the parameter interval. It gives a positive lower bound which persists throughout a neighborhood of the complex variable.

Lemma 19 (A uniform local bound)**.** *For every $z_0 \in D_\kappa$, there are $c > 0$ and $r > 0$ such that*

$$|1 + \kappa q z| \geq c \quad (q \in I,\ |z - z_0| < r).$$

Proof. The continuous positive function $q \mapsto |1 + \kappa q z_0|$ attains a strictly positive minimum a. Since $q \leq 1$, the reverse triangle inequality gives

$$|1 + \kappa q z| \geq a - \kappa |z - z_0|.$$

Take $c = a/2$ and $r = a/(2\kappa)$. The same estimate also proves that this ball lies in D_κ, so the domain is open. □

More generally, if K is a nonempty compact subset of D_κ, continuity on $K \times I$ gives

$$a_K := \min_{z \in K,\, q \in I} |1 + \kappa q z| > 0, \qquad \sup_{z \in K} |S_\kappa(z)| \leq \frac{\kappa M}{a_K}.$$

The first assertion uses positivity at every pair: compactness then excludes a sequence of denominators tending to zero. There is no need to estimate the distance to each point of the slit separately.

For continuity, choose a neighborhood with the common bound c and subtract the two kernels before integrating. The elementary identity

$$\frac{1}{1 + \kappa q z} - \frac{1}{1 + \kappa q w} = \frac{\kappa q (w - z)}{(1 + \kappa q z)(1 + \kappa q w)}$$

implies the local Lipschitz estimate

$$|S_\kappa(z) - S_\kappa(w)| \leq \frac{\kappa^2 M}{c^2} |z - w|.$$

Here both points lie in the chosen neighborhood. This proves continuity directly and supplies an explicit modulus there. Equivalently, dominated convergence applies because the kernels are bounded by c^{-1} and the constant function is integrable for a finite measure.

The same mechanism will handle higher derivatives. Powers of the coordinate remain bounded by one, while powers of the denominator are controlled by this single lower bound. The resulting majorants depend on the derivative order and the neighborhood, but not on the integration variable. This uniformity is the substantive analytic input in the compact-kernel argument formalized in the source development. It also controls perturbations of the measure. If τ is another finite positive measure and its transform is denoted $S_{\kappa,\tau}$, integration against the total variation of the signed difference gives

$$\sup_{z \in K} |S_{\kappa,\sigma}(z) - S_{\kappa,\tau}(z)| \leq \frac{\kappa}{a_K} |\sigma - \tau|(I).$$

Thus convergence in total variation implies uniform convergence on every compact subset of the common denominator domain.

3.3 Differentiation under the integral

Introduce the kernel tower and its integral by

$$K_n(z,q) = \frac{q^n}{(1+\kappa q z)^{n+1}}, \qquad A_n(z) = \int_I K_n(z,q)\, d\sigma(q), \qquad n \geq 0.$$

For each fixed parameter, ordinary complex differentiation gives

$$\partial_z K_n(z,q) = -(n+1)\kappa K_{n+1}(z,q).$$

The issue is to integrate this identity without relying on a purely formal differentiation rule. Fix $z_0 \in D_\kappa$ and a ball on which the denominator has the lower bound $c > 0$ from the preceding lemma. On that ball the relevant estimates are

$$|K_n(z,q)| \leq c^{-n-1}, \qquad |\partial_z K_n(z,q)| \leq (n+1)\kappa c^{-n-2}.$$

Both majorants are integrable because $M < \infty$. Measurability follows from continuity in q; it is not an additional assumption on the measure beyond its Borel structure.

Proposition 20 (Derivative of the kernel integral)**.** *Every A_n is holomorphic on D_κ, and*

$$A_n'(z) = -(n+1)\kappa A_{n+1}(z).$$

Proof. For nonzero complex h sufficiently small, the segment from z_0 to $z_0 + h$ stays in the chosen ball. Apply the real fundamental theorem of calculus to the smooth function of $t \in [0,1]$ obtained by restricting the rational kernel to this segment. It yields

$$\frac{K_n(z_0+h,q) - K_n(z_0,q)}{h} = \int_0^1 \partial_z K_n(z_0 + th, q)\, dt.$$

Consequently the difference quotient is bounded in modulus by the constant derivative majorant, uniformly in q and in small h. Pointwise in q it converges to $\partial_z K_n(z_0,q)$. Dominated convergence therefore moves this limit through the σ integral, proving complex differentiability and the stated derivative formula at z_0. Since z_0 was arbitrary and the domain is open, A_n is holomorphic there. □

This proof accommodates arbitrary directions of complex increments. It does not invoke a real mean-value theorem for complex-valued functions. It also explains why the derivative bound must hold on a neighborhood, rather than merely at the point where differentiation is performed. Repeating the proposition gives derivatives of every order and, by the usual complex-analytic theorem, convergent local Taylor expansions. These are the analytic germs needed for comparison with an actual logarithmic derivative. Atomic measures require no special exception: integration simply includes their point evaluations, and the same constant bounds apply. Nor does the proof need the measure to be a probability measure. Its total mass enters only through the integrals of the majorants, which are finite multiples of M.

3.4 The derivative tower and identification of germs

Write $u_n = \int_I q^n \, d\sigma(q)$ for the moments of the given measure. Since $S_\kappa = \kappa A_0$, induction using the preceding derivative formula proves

$$S_\kappa^{(n)}(z) = (-1)^n n! \, \kappa^{n+1} \int_I \frac{q^n}{(1+\kappa q z)^{n+1}} \, d\sigma(q).$$

Indeed, differentiation of the formula at order n multiplies the integral by $-(n+1)\kappa$ and increases its index by one, producing exactly the coefficient at order $n+1$. Evaluation at the origin gives

$$S_\kappa^{(n)}(0) = (-1)^n n! \, \kappa^{n+1} u_n.$$

There is also a direct convergence check: for $|\kappa z| < 1$, the geometric series of the kernel converges uniformly in $q \in I$. Its terms are bounded by the summable series $|\kappa z|^n$, so

$$S_\kappa(z) = \sum_{n=0}^{\infty} (-1)^n \kappa^{n+1} u_n z^n, \qquad |z| < 1/\kappa.$$

This radius is a guaranteed lower bound; a particular measure may give a larger radius.

Proposition 21 (An analytic identification criterion)**.** *If R is holomorphic near zero and its derivatives there are $R^{(n)}(0) = (-1)^n n! \kappa^{n+1} u_n$ for every n, then $R = S_\kappa$ on some neighborhood of zero.*

Proof. Choose a disk on which the Taylor expansion of R converges to R, and intersect it with $|z| < 1/\kappa$. Equality of the derivatives gives equality term by term of the two convergent Taylor series on this smaller disk, hence equality of their sums. □

For the actual application, set $\kappa = \texttt{actualKappa}$. The fixed function Θ has $\Theta(0) \neq 0$, so $R = \Theta'/\Theta$ is holomorphic near zero. The source's exact normalization and its retained moment-reality theorem provide the identity

$$R^{(n)}(0) = (-1)^n n! \, \kappa^{n+1} u_n^{\mathrm{act}}.$$

Thus a positive measure whose moments equal the actual normalized moments identifies the actual germ. The representability assertion is a mathematical hypothesis to be supplied by the preceding positivity argument; it is not obtained by defining a new sequence from an arbitrary measure. Likewise, agreement of finitely many moments gives only finitely many matching derivatives. The conclusion here requires all orders and analyticity, and therefore does not follow from a truncated computation or a merely formal coefficient list. For comparison, a finite truncation does have a rigorous error bound. Since $0 \le u_n \le M$, the geometric tail estimate yields

$$\left| S_\kappa(z) - \sum_{n=0}^{N} (-1)^n \kappa^{n+1} u_n z^n \right| \le \frac{\kappa M (\kappa|z|)^{N+1}}{1 - \kappa|z|}.$$

This bound controls the represented transform locally, but does not replace the exact all-orders comparison with the prescribed function.

3.5 The imaginary-part identity in two representations

Let $z = x + iy$ with $y > 0$. Conjugating the denominator gives the pointwise identity

$$\operatorname{Im} \frac{\kappa}{1 + \kappa q z} = -\frac{\kappa^2 q y}{|1 + \kappa q z|^2}.$$

Taking imaginary parts commutes with integration because it is a continuous real-linear map and the kernel is integrable. Consequently

$$\operatorname{Im} S_\kappa(z) = -\kappa^2 y \int_I \frac{q}{|1 + \kappa q z|^2}\, d\sigma(q) \le 0.$$

The sign follows from positivity of the measure and of the coordinate. It is strict precisely when $\sigma((0, 1]) > 0$: in that case the nonnegative integrand is positive on a set of positive measure. A measure concentrated at zero instead gives a real constant transform. Thus the nonpositive formulation includes all admissible measures.

The Hilbert-space representation explains the same calculation geometrically. On $H = L^2(I, \sigma; \mathbb{C})$ take $T = M_q$ and $v = 1$. We use inner products conjugate-linear in the first argument and linear in the second. Multiplication by the reciprocal denominator is bounded on D_κ, giving

$$S_\kappa(z) = \kappa \langle v, (I + \kappa z T)^{-1} v \rangle, \qquad \|v\|^2 = M.$$

The following algebra applies equally to any bounded positive selfadjoint contraction, whenever the displayed inverse exists. Put $h = (I + \kappa z T)^{-1} v$; then $v = h + \kappa z T h$, and

$$\langle v, h \rangle = \|h\|^2 + \kappa \overline{z} \langle Th, h \rangle, \qquad \langle Th, h \rangle = \langle h, Th \rangle \ge 0.$$

Multiplication by κ and extraction of the imaginary part yield

$$\operatorname{Im} S_\kappa(z) = -\kappa^2 y \langle h, Th \rangle.$$

In the multiplication model, the quadratic form on the right is exactly the integral in the scalar identity. The conjugate on z is forced by the inner-product convention; omitting it reverses the sign.

This is an anti-Herglotz convention: the upper half-plane is mapped into the closed lower half-plane. A resolvent written with $I - zT$ instead has the opposite sign. The concrete operator calculation is mathematical exposition of the measure model. Its validity does not by itself assert that every corresponding Hilbert-space packaging is an exported theorem of the accompanying Lean development. There is a complementary symmetry identity:

$$S_\kappa(\overline{z}) = \overline{S_\kappa(z)}.$$

Indeed, the parameters and the measure are real, so conjugation can be moved through the integral. This gives the corresponding nonnegative imaginary-part inequality in the lower half-plane. Both sign statements are consequences of one positive representation, with no assumption about zeros of a separate entire function.

3.6 Quantitative bounds and endpoint masses

The denominator estimates give useful controls without prescribing the singular behavior along the excluded slit. For $|z| \le r < 1/\kappa$, the reverse triangle inequality and the derivative tower give

$$|S_\kappa^{(n)}(z)| \le \frac{n!\,\kappa^{n+1} M}{(1-\kappa r)^{n+1}}, \qquad n \ge 0.$$

On the nonnegative real axis all denominators are positive. In particular, the transform is completely monotone there:

$$(-1)^n S_\kappa^{(n)}(x) = n!\,\kappa^{n+1} \int_I \frac{q^n}{(1+\kappa q x)^{n+1}}\, d\sigma(q) \ge 0, \qquad x \ge 0.$$

For $n = 0$ this also gives $S_\kappa(x) \le \kappa M$; if $M > 0$ the transform is strictly positive. The first derivative is nonpositive, although strict decrease fails when all mass is at zero.

Let $a_0 = \sigma(\{0\})$ and $a_1 = \sigma(\{1\})$. Splitting the measure into its endpoint atoms and its restriction to the open interval gives the exact decomposition

$$S_\kappa(z) = \kappa a_0 + \frac{\kappa a_1}{1+\kappa z} + \kappa \int_{(0,1)} \frac{d\sigma(q)}{1+\kappa q z}.$$

At positive real infinity, the kernel is bounded by one and converges pointwise to the indicator of $\{0\}$. Dominated convergence proves

$$\lim_{x \to +\infty} S_\kappa(x) = \kappa a_0.$$

No convergence rate follows merely from finiteness: mass may accumulate arbitrarily close to zero.

There is also a precise one-sided statement at the other endpoint. For $0 < \kappa\varepsilon \le 1$, the quantity $\kappa\varepsilon/(1 - q + \kappa q \varepsilon)$ lies between zero and one, because its denominator is at least $\kappa\varepsilon$. It tends to zero for $q < 1$ and equals one for $q = 1$. Hence

$$\lim_{\varepsilon \downarrow 0} \varepsilon S_\kappa(-1/\kappa + \varepsilon) = a_1.$$

This follows by dominated convergence along real points in D_κ. It identifies the endpoint atom without asserting meromorphic continuation across that endpoint. If additional mass accumulates at one, the remaining integral need not be holomorphic in any full neighborhood of $-1/\kappa$. Accordingly, the exact atomic summand and the proved one-sided limit must not be promoted to an unsupported claim that the entire transform has an isolated pole there. The disk bound is sharp uniformly over all measures: mass M at $q = 1$ attains equality at the negative real point $z = -r$. Likewise, the integral formula shows that every alternating derivative of positive order is strictly positive on the nonnegative real axis exactly when some mass lies away from zero. These observations identify the degenerate cases without imposing additional regularity assumptions on the measure.

3.7 Continuing an analytic defect from an origin germ

An identity of logarithmic derivatives near zero must be propagated without presupposing that the entire function has no zeros elsewhere. The appropriate object for the identity theorem is a holomorphic defect, defined without dividing by that function. Set

$$\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}.$$

Theorem 22 (Continuation of a differential identity)**.** *Let F be entire and let S be holomorphic on $\mathbb{H}$. Suppose that for some $\varepsilon > 0$ one has $F'(z) = S(z)F(z)$ whenever $z \in \mathbb{H}$ and $|z| < \varepsilon$. Then this identity holds throughout $\mathbb{H}$.*

Proof. The function

$$D(z) = F'(z) - S(z)F(z)$$

is holomorphic on $\mathbb{H}$. Although zero is not in this open half-plane, the hypothesized germ supplies an interior seed. Namely,

$$z_* = i\varepsilon/2 \in \mathbb{H} \cap B(0, \varepsilon).$$

A sufficiently small open disk about z_* lies in this intersection, and D vanishes on that disk. The upper half-plane is convex, hence connected. The identity theorem therefore gives $D = 0$ on all of $\mathbb{H}$, which is the required differential identity. □

To apply the theorem, suppose $F(0) \neq 0$ and its logarithmic derivative agrees near zero with a function S holomorphic both there and on $\mathbb{H}$. Continuity allows the germ neighborhood to be shrunk so that F has no zeros in it. Multiplication by F then gives

$$\frac{F'}{F} = S \quad \text{near } 0 \quad \Longrightarrow \quad F' = SF \quad \text{on } \mathbb{H}.$$

Only after continuation do we divide at individual points where the denominator is known to be nonzero:

$$F(z) \neq 0,\ z \in \mathbb{H} \quad \Longrightarrow \quad \frac{F'(z)}{F(z)} = S(z).$$

Thus any imaginary-part bound for S transfers to the logarithmic derivative at precisely the points needed for pole exclusion.

For the measure resolvent, holomorphy on $\mathbb{H}$ follows from $\mathbb{H} \subset D_\kappa$. For the actual entire function, the previous germ criterion supplies the local equality when its moments are represented by σ. The argument keeps these two inputs separate: positivity constructs and controls S, while analyticity and exact moment agreement identify the germ. Global zero-freeness is not one of the inputs to this continuation step. The defect identity itself remains meaningful at a hypothetical zero. At such a point it states $F' = 0$, which by itself would only exclude a simple zero. The stronger conclusion comes from the identity on a whole neighborhood and the local behavior of logarithmic derivatives, examined next.

3.8 The pole probe and conjugate-zero symmetry

The sign constraint has a local consequence for any nontrivial entire function. Its proof uses the positive integral multiplicity of a holomorphic zero, independently of how the sign constraint was obtained.

Theorem 23 (Upper-half-plane pole obstruction)**.** *Let F be entire and not identically zero. If*

$$\operatorname{Im}\frac{F'(z)}{F(z)} \le 0 \quad \textit{for every } z \in \mathbb{H} \textit{ with } F(z) \neq 0,$$

then F has no zeros in $\mathbb{H}$.

Proof. Suppose $\rho \in \mathbb{H}$ is a zero. Local factorization gives a positive integer m, a disk about ρ, and a holomorphic function g without zeros on that disk such that

$$F(z) = (z-\rho)^m g(z), \qquad \frac{F'(z)}{F(z)} = \frac{m}{z-\rho} + B(z), \quad B = \frac{g'}{g},$$

where the quotient identity holds away from ρ. On a smaller closed disk, continuity of B gives a finite bound $|\operatorname{Im} B| \le K$ with $K \ge 0$. Choose $\varepsilon > 0$ small enough that $\rho - i\varepsilon$ lies in this smaller disk, $\varepsilon < \operatorname{Im}\rho$, and $\varepsilon < m/(K+1)$. The probe point is in $\mathbb{H}$ and is not a zero, because the factor g does not vanish. At that point,

$$\operatorname{Im}\frac{F'(\rho-i\varepsilon)}{F(\rho-i\varepsilon)} = \frac{m}{\varepsilon} + \operatorname{Im} B(\rho-i\varepsilon) \ge \frac{m}{\varepsilon} - K > 0.$$

This contradicts the assumed sign inequality. □

The downward direction is decisive: the imaginary part of $m/(-i\varepsilon)$ is positive. An upward probe would give the opposite sign and would not establish this contradiction. The local factorization also ensures that the tested quotient is defined; there is no appeal to a value at the putative pole itself.

If the zeros are invariant under conjugation, the lower half-plane is excluded as well. Indeed,

$$F(\rho) = 0 \Longrightarrow F(\overline{\rho}) = 0, \qquad \operatorname{Im}\rho < 0 \Longrightarrow \operatorname{Im}\overline{\rho} > 0.$$

A sufficient condition is Schwarz symmetry $F(\overline{z}) = \overline{F(z)}$, although only the implication for zeros is needed. Combining this symmetry with the theorem leaves real zeros as the only possibility. Applied after the analytic-defect continuation, this is the general mechanism by which a positive measure representation constrains the zeros of the actual entire function. The next chapter specifies the actual-function bindings needed to turn that constraint into the repository's stated endpoint. The limitation to nonreal zeros is necessary. For example, if a is real, $F(z) = z - a$ satisfies the sign hypothesis because

$$\operatorname{Im}\frac{1}{z-a} = -\frac{\operatorname{Im} z}{|z-a|^2} < 0 \qquad (z \in \mathbb{H}),$$

yet has a real zero at a. The theorem therefore imposes exactly the stated location constraint.

4.1 The actual Riemann xi function

The application requires a fixed analytic function, including its values at the apparent singularities of the usual completion. Write Λ_0 for Mathlib's entire pole-subtracted completion, named `completedRiemannZeta₀` in the mathematical description. The source definition is exactly

$$\xi(s) = \frac{1 + s(s-1)\Lambda_0(s)}{2}, \qquad s \in \mathbb{C}. \tag{12}$$

This formula defines an entire function without introducing removable values by convention. At the two exceptional arguments it immediately gives

$$\xi(0) = \xi(1) = \tfrac{1}{2}. \tag{13}$$

If Λ denotes the meromorphic completed zeta function, the retained comparison theorem identifies the familiar expression on its valid domain:

$$\xi(s) = \frac{s(s-1)}{2}\Lambda(s) \qquad (s \neq 0, 1). \tag{14}$$

Keeping these domains explicit prevents a totalized division operation in the formal language from being mistaken for analytic evaluation at a pole. It also fixes the normalization used later for the logarithmic derivative.

Proposition 24 (Retained literal RH binding)**.** *For this particular ξ, Mathlib's proposition `RiemannHypothesis` is equivalent to*

$$\forall s \in \mathbb{C}, \qquad \xi(s) = 0 \implies \operatorname{Re} s = \tfrac{1}{2}. \tag{15}$$

The source proves this equivalence by matching the zeros of ξ with the nontrivial zeros in Mathlib's definition. The latter excludes the negative even integers and the exceptional argument 1. The factors in the completion are checked at the relevant arguments; the nonvanishing theorems also rule out extraneous zeros. Thus the endpoint concerns the ordinary Riemann hypothesis, rather than a proposition chosen to resemble it or a model zero predicate.

This binding is a retained theorem from `ActualXi.lean`, and the entire normalization comes from `AWPIRH/XiCore.lean` [7]. We do not reprove the completion and nonvanishing infrastructure here. Subsequent implications always return to this same proposition. In particular, changing a moment convention or constructing an abstract positive operator cannot silently change the analytic function to which the conclusion applies. The role of the following pages is to exhibit the exact connections that make such a return legitimate. An overall nonzero constant would leave the logarithmic derivative unchanged, but retaining the exact normalization makes the source correspondence unambiguous throughout.

4.2 Entire descent through the centered square

The reflection equation of the actual completion places the center of symmetry at one half. Introduce the centered function F and the descended function Θ by

$$\xi(1-s)=\xi(s), \qquad F(w)=\xi(\tfrac{1}{2}+w), \qquad F(-w)=F(w), \tag{16}$$

$$\Theta(z)=F(\sqrt{z}), \qquad \Theta(w^2)=F(w). \tag{17}$$

Here the square root in the definition is a chosen pointwise square root. The second identity is independent of that choice: the two roots differ only by sign, and F is even. Neither this observation nor the notation asserts that a global holomorphic square root exists.

Proposition 25 (Retained entire descent)**.** *The function Θ is entire, satisfies* (17)*, and has $\Theta(0)\neq 0$.*

For completeness, the analytic mechanism can be described locally. Near a nonzero point one selects a holomorphic square root after normalizing the argument into a neighborhood of 1. Evenness identifies the resulting composition with the pointwise definition. At zero, continuity and the removable-singularity theorem complete the extension. Equivalently, an even Taylor expansion descends by replacing w^{2n} with z^n. The source implements the local-root and removable-singularity route in `ThetaEntire.lean`; origin nonvanishing is supplied separately.

The actual-function development also supplies conjugation symmetry:

$$\Theta(\overline{z})=\overline{\Theta(z)}, \qquad \Theta(z)=0 \implies \Theta(\overline{z})=0. \tag{18}$$

Its implications differ from those of evenness. Evenness identifies the two centered xi arguments over a single square; conjugation pairs the upper and lower half-plane zeros of the descended function. Neither property alone places all zeros on a line.

The exact geometric translation is

$$\mathrm{RH} \iff \big[\Theta(z)=0 \implies z\in(-\infty,0]\big]. \tag{19}$$

Indeed, a centered critical-line point has the form it, whose square is $-t^2$. Conversely, a square is real and nonpositive precisely when its square root has zero real part. Applying the centered-square identity in both directions proves the translation, using the literal RH binding already fixed. Origin nonvanishing subsequently removes the endpoint zero from the actual divisor. Away from that endpoint, the square map has nonzero derivative. Local zero orders are therefore preserved when descending either member of an opposite pair. Pairing those members avoids counting the same descended zero twice.

4.3 Exact moments and their positive normalization

Origin nonvanishing gives a disk on which the actual logarithmic derivative is holomorphic. Define the complex moments before taking real parts:

$$R(z) = \frac{\Theta'(z)}{\Theta(z)}, \qquad a_n = \frac{(-1)^n}{n!} R^{(n)}(0), \qquad m_n = \operatorname{Re} a_n. \tag{20}$$

The real coordinate in this definition is convenient for Bernstein rows, but does not discard an unaccounted imaginary part. The retained reality theorem proves $a_n = \overline{a_n}$ at every order. Schwarz reflection of Θ passes to its derivative, quotient, and iterated derivatives; evaluation at the real point zero therefore yields

$$a_n = m_n \in \mathbb{R} \qquad (n \geq 0). \tag{21}$$

This argument uses neither RH nor a preliminary assertion that the canonical nodes are real. It is also independent of the later all-order power-sum theorem. These distinctions are essential when the moments are used to prove a zero-location criterion.

Write A_i for the canonical complex scales of the paired actual divisor, with multiplicity. The source supplies the absolutely convergent first trace and its positivity:

$$\kappa := m_0 = \sum_i \operatorname{Re}(A_i^{-1}) > 0. \tag{22}$$

The proof uses the actual product, unconditional low-height zero exclusion, and nonempty divisor support. Specifically, the scale geometry gives $\operatorname{Re}(A_i^{-1}) > 0$ for every index. Summability permits comparison with one strictly positive term, proving strict positivity of the total. It does not assume that every scale is positive real; that stronger conclusion would already encode the desired zero geometry.

The normalized moments and their derivative calibration are consequently

$$u_n = \frac{m_n}{\kappa^{n+1}}, \qquad u_0 = 1, \qquad R^{(n)}(0) = (-1)^n n! \kappa^{n+1} u_n. \tag{23}$$

All divisions here are justified by the established strict positivity. The sequence u is now fixed by the actual analytic function, rather than selected to satisfy a moment criterion. The accompanying modules `ActualMomentReality.lean` and `ActualCalibrationProgress.lean` supply the retained inputs [7]. They discharge reality and normalization before either forward route assumes H_{00}. The remaining question is whether this particular real sequence has positive Bernstein rows at every order. Reality itself supplies no such positivity: a real sequence can have negative moments or negative finite differences. The alternating sign convention fixes precisely which differences the later criterion tests.

4.4 Canonical nodes and exact power sums

The paired divisor turns the moment sequence into spectral data without presupposing a positive measure. With the complex scales A_i from the preceding page, define

$$q_i = (\kappa A_i)^{-1}, \qquad \alpha_i = -A_i = -\frac{1}{\kappa q_i}. \tag{24}$$

The canonical index retains repeated occurrences, so zero multiplicities are preserved. Its calibration includes both inclusion and exhaustion:

$$q_i \neq 0, \quad \Theta(\alpha_i) = 0, \qquad \Theta(z) = 0 \implies \exists i \ (z = \alpha_i). \tag{25}$$

Exhaustion is indispensable. A family consisting only of favorable zeros could establish neither the spectral criterion nor the return to RH. The source obtains it from the entire actual divisor and its opposite-pair indexing, not from a finite list of numerically located zeros.

Theorem 26 (Retained canonical power sums)**.** *The actual nodes satisfy*

$$\sum_i |q_i| < \infty, \qquad u_n = \sum_i q_i^{n+1} \quad (n \geq 0), \qquad \sum_i q_i = 1. \tag{26}$$

The sums are convergent complex sums; their stated values are real.

The final mass identity is the case $n = 0$. It does not imply positivity of individual nodes, since cancellation remains possible in a complex sum. Absolute summability, however, bounds each node and controls every higher power, supplying the necessary convergence statements.

Here is the analytic content behind the all-order identification. Put $b_i = A_i^{-1}$ and $M = 1 + \sum_i |b_i|$. On a common neighborhood of zero,

$$|z|M < \tfrac{1}{2} \quad \implies \quad |1 + zb_i| > \tfrac{1}{2}, \qquad \left|\frac{b_i}{1 + zb_i}\right| \leq 2|b_i|. \tag{27}$$

The same summable family, multiplied by a constant depending on the derivative order, majorizes every resolvent power. Locally uniform product convergence identifies the logarithmic derivative with the resolvent sum; holomorphic derivative interchange then produces all power sums. Reality and the exact κ normalization finish the calculation.

The module `CanonicalPowerSums.lean` supplies this construction; `AnalyticClosure.lean` packages its fields as `actualCanonicalCalibration`. Thus calibration is constructed analytic data at the final endpoints. It remains an infinite mathematical object, and its existence is neither a finite certificate nor a proof of H_{00}. Derivative interchange is justified separately at each finite order on the same open neighborhood. Its majorant may depend on that order; no uniform bound over all derivative orders is asserted or needed for these identities.

4.5 Ellipse geometry and the spectral radius

For a complex node q, the quantity $|q| + |1 - q|$ measures its position relative to the two real foci zero and one. It is at least one by the triangle inequality. Equality requires the two summands to have compatible directions, giving the elementary characterization

$$E(q) := |q| + |1 - q| \geq 1, \qquad E(q) = 1 \iff q \in [0, 1]. \tag{28}$$

The interval is regarded as a subset of $\mathbb{C}$. In particular, real nodes outside the interval and genuinely nonreal nodes both give strictly larger values. This geometry detects more than the imaginary part alone. For a fixed value greater than one, the level set is an ellipse with those two foci. Thus the radius describes the largest ellipse reached by the calibrated nodes. It is a radius for this affine Bernstein geometry, not the usual supremum of the node moduli. Replacing it by that simpler supremum would lose the contribution from the factor involving one minus the node.

Define the radius of the actual calibrated family by

$$\rho = \sup\big(\{1\} \cup \{E(q_i) : i \in J\}\big). \tag{29}$$

The inserted value one makes the lower bound explicit. Absolute summability makes the family bounded, since $E(q_i) \leq 1 + 2|q_i|$. Consequently the supremum is finite, and

$$\rho = 1 \quad \iff \quad \forall i \ (q_i \in [0, 1]). \tag{30}$$

For the actual calibration, this is exactly RH. If the nodes lie in the interval, nonvanishing makes them positive; every calibrated zero $-1/(\kappa q_i)$ is then negative real. Exhaustion and (19) give RH. Conversely, RH places all descended zeros on the negative axis and therefore makes each node positive real. The identity $\sum_i q_i = 1$ bounds each by one.

The connection with row variation is visible in the exact generating polynomial. For $|x| \leq 1$, absolute summability justifies the formula

$$P_N(x) := \sum_{j=0}^{N} w_{N,j} x^j = \sum_i q_i (1 - q_i + x q_i)^N, \qquad |P_N(x)| \leq V_N. \tag{31}$$

Taking absolute values before summing produces the useful bound

$$1 \leq V_N \leq \sum_i |q_i| E(q_i)^N \leq \left(\sum_i |q_i| \right) \rho^N. \tag{32}$$

This proves the upper exponential rate immediately. Establishing the matching lower rate requires a separate argument, because complex contributions to the generating polynomial can cancel. The next page states precisely the retained theorem that resolves this issue.

4.6 The retained ordinary spectral limit

Theorem 27 (Retained actual spectral closure)**.** *For the actual normalized sequence and its canonical nodes,*

$$\lim_{N\to\infty} V_N^{1/N} = \rho. \tag{33}$$

This is an ordinary limit along positive integers.

The source constructs the required dominant extraction from the calibration and then applies its spectral limit theorem. We retain that result here; the following description identifies its obligations without claiming to reproduce the full extraction proof. The case $\rho = 1$ follows directly from (32). Suppose therefore that $\rho > 1$.

Summability forces the nodes to approach zero outside finite sets, so their ellipse radii approach one there. A maximizing node can therefore be isolated in a finite region. Choosing a unit complex phase x aligns its two affine contributions. At this phase the row polynomial admits the constructed decomposition

$$P_N(x) = \sum_{\ell=1}^{d} c_\ell \lambda_\ell^N + T_N, \qquad |x| = 1, \quad |\lambda_\ell| = \rho. \tag{34}$$

The rates are distinct after grouping repeated nodes; at least one coefficient is nonzero. Equal nodes have coefficients equal to their positive integer multiplicity times the node, so grouping cannot erase every dominant coefficient. The complement has a uniform strict bound:

$$|T_N| \le C r^N, \qquad 1 \le r < \rho. \tag{35}$$

This complement includes both the infinite tail and the finite nodes whose affine rate is smaller than the maximum.

Distinct dominant phases can cancel at particular indices. The argument therefore uses finite Vandermonde estimates over blocks of consecutive indices, rather than asserting a pointwise lower bound for every exponential sum. The strict tail estimate makes the remainder negligible. The inequalities $|P_N(x)| \le V_N$ and monotonicity of V_N transfer the block lower estimates to the ordinary root limit. Those are the steps for which the retained spectral infrastructure is needed.

Combining the limit with (30) yields

$$\left[\forall \varepsilon > 0\ \exists A > 0\ \forall N \ge 0,\ V_N \le A e^{\varepsilon N}\right] \quad \Longleftrightarrow \quad \text{RH}. \tag{36}$$

The exact source declarations are `canonical_actual_root_tendsto_constructed` and `canonical_actual_subexponential_iff_nativeRH` in `AnalyticClosure.lean` [7]. For the forward implication in this last criterion, each exponential bound forces the limit to be at most its exponential base. Letting that base decrease to one gives the required radius.

4.7 The first route and the shared converse

The first route now combines finite cancellation algebra with the retained actual spectral criterion. Recall that H_{00} is semantic admissibility of the existing cutoff-zero, coefficient-zero budget. Its meaning includes every order; the finite notation for the candidate does not restrict that quantifier. For the actual rows, the identities proved earlier give

$$H_{00} \implies C_n = 0 \ (n \geq 0) \implies V_{n+1} = V_n \ (n \geq 0). \tag{37}$$

The second implication follows from $C_n = (n+1)(V_{n+1} - V_n)$ and the positivity of $n+1$. Starting with the unconditional normalization $V_0 = 1$, induction yields

$$H_{00} \quad \Longrightarrow \quad V_n = 1 \qquad (n \geq 0). \tag{38}$$

For each $\varepsilon > 0$, choosing $A = 1$ supplies the required subexponential estimate:

$$V_n = 1 \leq e^{\varepsilon n} \qquad (n \geq 0). \tag{39}$$

Equation (36) therefore proves the first implication $H_{00} \Rightarrow$ RH. Its mathematical content beyond finite row algebra is exactly the constructed actual spectral closure. No separate extraction or calibration hypothesis remains in the endpoint. The proof also identifies the logical order: row positivity is available directly from zero cancellation before the spectral implication is invoked. The second route may therefore reuse that positivity without assuming the first route's conclusion.

The reverse implication is shared by both routes. Assume RH. The actual calibration and radius geometry place each q_i in $[0, 1]$. Substituting the power sums into the finite Bernstein difference formula gives

$$w_{N,j} = \binom{N}{j} \sum_i q_i^{j+1} (1 - q_i)^{N-j} \geq 0, \qquad 0 \leq j \leq N. \tag{40}$$

Absolute summability justifies exchanging the infinite node sum with the finite difference sum. Each resulting term is nonnegative. The row mass identity then gives $V_N = 1$, hence zero cancellation and H_{00}.

We have consequently obtained the equivalence

$$H_{00} \quad \Longleftrightarrow \quad \text{RH}. \tag{41}$$

The common reverse construction is exposed in the source as `H00_of_nativeRH`. The equivalence does not discharge its left-hand assertion: it characterizes RH by another universal statement. In particular, checking any fixed collection of rows cannot supply the induction hypothesis used above for all later degrees. A further theorem would be needed to propagate finite data to every cancellation order. This unresolved issue is separate from the validity of the two conditional implications.

4.8 The second route and the actual RH endpoint

The second forward construction applies the preceding measure–resolvent argument to the actual moments. From H_{00}, finite row algebra gives positive normalized Bernstein rows. The Hausdorff construction produces a probability measure σ on $[0,1]$, whose moments are exactly u_n. Its transform is

$$S(z) = \kappa \int_{[0,1]} \frac{d\sigma(q)}{1+\kappa q z}, \qquad \operatorname{Im} S(z) \le 0 \quad (\operatorname{Im} z > 0). \tag{42}$$

The actual derivative calibration (23) identifies the analytic germs of S and R at zero. Continuing the holomorphic defect, rather than dividing at possible zeros, gives

$$\Theta' - S\Theta = 0 \quad \text{on the upper half-plane.} \tag{43}$$

An upper-half-plane zero a of multiplicity $m \ge 1$ would have $R(z) = m/(z-a) + B(z)$ with B holomorphic nearby. At $z = a - i\varepsilon$ the principal part has imaginary part $m/\varepsilon > 0$, contradicting the sign after the bounded remainder is absorbed. Conjugate-zero symmetry then makes every Θ-zero real.

For an actual xi zero $s = \sigma_0 + it$, centered-square descent therefore implies

$$0 = \operatorname{Im}(s - \tfrac{1}{2})^2 = 2(\sigma_0 - \tfrac{1}{2})t, \qquad |t| > 1. \tag{44}$$

The last inequality is a retained unconditional zero-height theorem. It forces $\sigma_0 = 1/2$, and Proposition 24 gives literal RH. This closure needs no separate exclusion of positive-real Θ-zeros. Every analytic input is either constructed from H_{00} or supplied for the fixed actual functions. In particular, the measure is not posited independently of the input sequence. Exact moment identification constructs the origin germ, and origin nonvanishing suffices there. The continuation step is formulated with an analytic defect precisely so that hypothetical zeros elsewhere do not become unmentioned assumptions in the proof.

In namespace `RH_HilbertLimit.TwoRoutes`, the four exact exported declarations are

```
nativeRH_of_H00_route_one
H00_iff_nativeRH_route_one
nativeRH_of_H00_route_two
H00_iff_nativeRH_route_two.
```

Their types are respectively $H00 \to$ `RiemannHypothesis` and $H00 \leftrightarrow$ `RiemannHypothesis` for each route; the formalization appendix records their verification context. The second forward proof does not call the first RH endpoint, although both share foundations and the reverse implication. The explanatory L^2 operator model introduces no additional audited Lean result. Neither endpoint proves H_{00} unconditionally or constructs a terminating RH checker.

5.1 Normalized weights and contraction multipliers

Fix a probability measure σ on $[0,1]$, a scale $\kappa > 0$, and an anchor $a > 0$. In the arithmetic application, these are the actual scale and the Hausdorff measure supplied under H_{00}. The constructions on this page also make sense for any such measure; their elementary inequalities do not identify its moments with a prescribed analytic function. Define

$$w_a(q) = \frac{4\kappa a}{(1+\kappa a q)^2}, \qquad \lambda_a(q) = q w_a(q) = \frac{4\kappa a q}{(1+\kappa a q)^2}. \tag{45}$$

These continuous functions separate the positive density from the variable which will act by multiplication. This distinction matters: the weight need not be bounded by one, whereas the multiplier always is.

Proposition 28. *For every $q \in [0,1]$,*

$$0 < w_a(q) \le 4\kappa a, \qquad 0 \le \lambda_a(q) \le 1. \tag{46}$$

Moreover, $\lambda_a(q) = 1$ if and only if $\kappa a q = 1$.

Proof. Set $t = \kappa a q \ge 0$. The denominator $(1+t)^2$ is positive and at least one, proving the weight bounds. The multiplier is nonnegative, and its upper bound and equality condition follow from the exact identity

$$1 - \lambda_a(q) = \frac{(1-t)^2}{(1+t)^2}. \tag{47}$$

Thus equality is possible on the support interval only when $\kappa a \ge 1$, and then only at $q = (\kappa a)^{-1}$. □

For every integer $n \ge 0$, multiplication by the nonnegative weight therefore gives the pointwise domination

$$0 \le w_a(q)\lambda_a(q)^n \le w_a(q) \le 4\kappa a. \tag{48}$$

This bound is uniform in the order n and in q. It needs neither a density for σ nor the exclusion of endpoint atoms. In particular, the zero endpoint has the explicit values

$$w_a(0) = 4\kappa a, \qquad \lambda_a(0) = 0, \qquad w_a(0)\lambda_a(0)^n = 0 \quad (n \ge 1). \tag{49}$$

An atom at zero contributes to the zeroth coefficient and disappears at every positive order. Keeping that contribution is necessary when comparing the base energy with later moments. Conversely, strict positivity of the weight implies that its integral against a probability measure is strictly positive. Thus the base vector constructed below cannot vanish, even when the entire measure is concentrated at an endpoint. If $\kappa a < 1$, compactness even gives $\sup_q \lambda_a(q) < 1$; however, the unit bound is the statement valid uniformly over all positive anchors. The basic kernel inequalities are audited in `RouteTwoContractive` and `ActualPositivePrefix`; the equality case and the following Hilbert interpretation are paper proofs [7].

5.2 A concrete weighted-vector realization

Work in the complex Hilbert space $\mathcal{H} = L^2([0,1], \sigma)$, with inner product linear in its second argument. Put

$$v_a = \sqrt{w_a}, \qquad (T_a f)(q) = \lambda_a(q) f(q), \qquad \widetilde{\mu}_n(a) = \int_{[0,1]} w_a(q) \lambda_a(q)^n \, d\sigma(q). \tag{50}$$

The tilde distinguishes these measure coefficients from the actual Widder tower. Since $w_a \le 4\kappa a$ and σ has mass one, v_a belongs to $\mathcal{H}$. Multiplication respects equality almost everywhere, so T_a is well defined on the equivalence classes defining L^2.

Proposition 29. *The operator T_a is a bounded positive self-adjoint contraction, and*

$$\widetilde{\mu}_n(a) = \langle v_a, T_a^n v_a \rangle, \qquad 0 \le \widetilde{\mu}_n(a) \le \widetilde{\mu}_0(a) \le 4\kappa a. \tag{51}$$

Proof. The range bound on λ_a gives

$$\|T_a f\|^2 = \int \lambda_a^2 |f|^2 \, d\sigma \le \|f\|^2, \qquad \langle f, T_a f \rangle = \int \lambda_a |f|^2 \, d\sigma \ge 0. \tag{52}$$

Because λ_a is real, moving it between the two inner-product arguments proves self-adjointness. Induction gives $T_a^n f = \lambda_a^n f$. Substitution of v_a proves the coefficient identity, and integration of the preceding page's pointwise bounds proves all inequalities. □

The coefficient estimate also follows from a decomposition that survives in much weaker geometry. For $v = v_a$ and $x = T_a^n v_a$, write

$$\begin{aligned} \|v\|^4 - |\langle v, x \rangle|^2 = {} & \underbrace{\|v\|^2 (\|v\|^2 - \|x\|^2)}_{D_{\mathrm{c}}} \\ & + \underbrace{\|v\|^2 \|x\|^2 - |\langle v, x \rangle|^2}_{D_{\mathrm{g}}}. \end{aligned} \tag{53}$$

Contractivity makes D_{c} nonnegative; Cauchy–Schwarz does the same for D_{g}. Thus the defect is the squared Euclidean norm of $(\sqrt{D_{\mathrm{c}}}, \sqrt{D_{\mathrm{g}}})$. No spectral theorem is needed.

The integral formula supplies more than individual coefficient bounds. For any polynomial $P(t) = \sum_{j=0}^{d} c_j t^j$,

$$\sum_{i,j=0}^{d} \overline{c_i} c_j \widetilde{\mu}_{i+j}(a) = \int w_a |P(\lambda_a)|^2 \, d\sigma \ge 0. \tag{54}$$

Replacing the weight by $w_a(1 - \lambda_a)$ similarly proves positivity of the matrix of successive moment differences. Both assertions follow by expanding a finite sum under the integral. They describe additional structure of this particular realization, beyond the single-coefficient estimate required at each order by the abstract orbit argument.

The audited `RealGramGeometry` and `BKQRContractiveOrbit` interfaces retain precisely enough scalar inequalities for the real version of this argument. Their carrier has no required vector-space structure, and their map T is merely a function with a norm-square inequality. In particular, those interfaces do not assert linearity, self-adjointness, completeness, or existence of this L^2 model. The audited defect-square theorem applies to a supplied realization; the concrete realization above is an additional mathematical construction [7].

5.3 The coefficient resolvent on the unit disk

The concrete contraction makes a generating function available before any identification with the arithmetic Widder tower. For a complex parameter h with $|h| < 1$, the operator series converges in norm and gives

$$(I - hT_a)^{-1} = \sum_{n=0}^{\infty} h^n T_a^n, \qquad \|(I - hT_a)^{-1}\| \le \frac{1}{1 - |h|}. \tag{55}$$

Indeed, the operator norm of its tail is bounded by a geometric tail, and multiplication by $I - hT_a$ telescopes. The same inverse is multiplication by $(1 - h\lambda_a)^{-1}$, since $|1 - h\lambda_a(q)| \ge 1 - |h|$ everywhere. Consequently

$$G_a(h) := \langle v_a, (I - hT_a)^{-1} v_a \rangle = \int \frac{w_a(q)}{1 - h\lambda_a(q)}\, d\sigma(q) = \sum_{n=0}^{\infty} \widetilde{\mu}_n(a) h^n. \tag{56}$$

Proposition 30. *The function G_a is holomorphic on the unit disk. For $0 \le r < 1$ and $|h| \le r$,*

$$\begin{aligned} |G_a(h)| &\le \frac{\widetilde{\mu}_0(a)}{1 - r}, \\ \left| G_a(h) - \sum_{n=0}^{N} \widetilde{\mu}_n(a) h^n \right| &\le \frac{\widetilde{\mu}_0(a) r^{N+1}}{1 - r}. \end{aligned} \tag{57}$$

For every integer $k \ge 0$, its derivatives satisfy

$$G_a^{(k)}(h) = k! \int \frac{w_a \lambda_a^k}{(1 - h\lambda_a)^{k+1}}\, d\sigma, \qquad |G_a^{(k)}(h)| \le \frac{k! \widetilde{\mu}_0(a)}{(1 - r)^{k+1}}. \tag{58}$$

Proof. The coefficient bound from the preceding page supplies a summable geometric majorant on every closed subdisk. Uniform convergence of holomorphic partial sums proves holomorphy and the stated remainder estimate. To differentiate the integral, choose s with $r < s < 1$. Each differentiated integrand on $|h| \le s$ is bounded by $k! w_a/(1 - s)^{k+1}$, an integrable function. Dominated differentiation therefore applies. Evaluating the resulting integrand on $|h| \le r$ gives the sharper displayed bound. □

These estimates are uniform for anchors in any compact positive interval $[\alpha, \beta]$, because $\widetilde{\mu}_0(a) \le 4\kappa\beta$. They establish a stable analytic object with controlled truncation error. They give no boundary convergence assertion at $|h| = 1$, where an atom with $\lambda_a = 1$ contributes a term proportional to $(1 - h)^{-1}$. If $\lambda_a \le \rho < 1$ almost everywhere, the same proof instead extends the series to $|h| < \rho^{-1}$ when $\rho > 0$. When $\rho = 0$, the function is constant. Thus the unit disk is a uniform guarantee, not a claim that every model has radius exactly one. Coefficient extraction in that disk gives

$$G_a^{(n)}(0) = n!\, \widetilde{\mu}_n(a). \tag{59}$$

This recovers the represented moments uniquely from the constructed analytic function and fixes their normalization without an arbitrary scalar factor. Finally, h is the coefficient-series variable; it is not automatically the variable in an actual angular observable. Such an interpretation requires a separately proved coefficient identity, to which we now turn.

5.4 The explicit actual-coefficient obligation

The archived actual tower is defined from the literal logarithmic derivative, rather than from a measure. Write $R(a)$ for `actualRealLogDerivative` and $\mu_n^{\mathrm{act}}(a)$ for `actualWidderMoment`. Its definitions are

$$B(a) = 4a\frac{d}{da}(aR(a)), \qquad \mu_0^{\mathrm{act}} = B, \qquad \mu_{n+1}^{\mathrm{act}} = \mathcal{W}_n \mu_n^{\mathrm{act}}, \tag{60}$$

where, with $m = n + 1$, the specified differential operator is

$$(\mathcal{W}_n f)(a) = \frac{2}{m(2m+1)}\left(m^2 f(a) - af'(a) - a^2 f''(a)\right). \tag{61}$$

These definitions must be retained when stating a representation theorem. Replacing their right-hand sides by positive integrals would change the object requiring identification.

Here κ is the actual positive scale. Let σ_N denote the finite Bernstein measure of the actual normalized origin moments, using the repository's grid convention. Under H_{00} it is a probability measure. The finite quantity called `actualKernelApproximation` is

$$A_{N,n}(a) = \int w_a(q)\lambda_a(q)^n \, d\sigma_N(q). \tag{62}$$

The named proposition `ActualApproximationConvergence` is exactly the all-order, all-positive-anchor assertion

$$\forall a > 0 \; \forall n \in \mathbb{N}, \qquad \lim_{N\to\infty} A_{N,n}(a) = \mu_n^{\mathrm{act}}(a). \tag{63}$$

The source exposes this as a hypothesis; it supplies no inhabitant. For comparison, weak convergence $\sigma_N \to \sigma$ identifies the same limit with $\widetilde{\mu}_n(a)$, because the kernel is continuous on the compact support interval. Thus identifying the actual differential tower, rather than merely constructing a measure, is the decisive step. Uniform bounds on the values of the finite approximants do not, by themselves, justify passage of first or second derivatives through a limit. Any argument using the defining recursion must establish the appropriate regularity and differentiated convergence separately.

Proposition 31. *Assume H_{00} and* (63)*. Then, for $a > 0$,*

$$0 \le \mu_n^{\mathrm{act}}(a) \le \mu_0^{\mathrm{act}}(a), \qquad \sum_{n=0}^{\infty} |\mu_n^{\mathrm{act}}(a)| r^n < \infty \quad (0 \le r < 1). \tag{64}$$

Proof. The finite bounds $0 \le A_{N,n}(a) \le A_{N,0}(a)$ hold at every depth. Passing both sides to their assumed limits preserves order. The resulting coefficient bound gives the geometric majorant $\mu_0^{\mathrm{act}}(a)\sum_n r^n$. □

These conditional consequences are audited in `ActualLimitBridge` [7]. They do not discharge the original longer angular/Widder identification program. The completed measure-resolvent route uses exact origin moments and an analytic defect directly; it reaches its conditional $H_{00} \Rightarrow \mathrm{RH}$ endpoint without this additional coefficient obligation. The distinction prevents a completed route from being mistaken for a proof of every proposed intermediate construction.

5.5 Finite data and universal admissibility

A finite pair of integers can name a mathematical condition that is not itself a finite computation. The repository's cancellation budget is precisely such a pair. With C_n the actual row cancellation and V_n the actual variation, its data and validity are

$$W = (N_0, K) \in \mathbb{N}^2, \qquad \mathrm{Adm}(W) :\Longleftrightarrow \forall n \geq N_0,\ C_n \leq K V_n. \tag{65}$$

The word "finite" describes the data fields, whereas admissibility quantifies over an infinite tail. In particular, nonnegativity of C_n makes the distinguished zero budget equivalent to

$$H_{00} = \mathrm{Adm}(0,0) \quad \Longleftrightarrow \quad \forall n \geq 0,\ C_n = 0. \tag{66}$$

The proved semantic witness theorem is

$$\big(\exists W \in \mathbb{N}^2,\ \mathrm{Adm}(W)\big) \quad \Longleftrightarrow \quad \mathrm{RH}. \tag{67}$$

Its forward implication uses the cancellation identity and eventual growth bounds; its reverse implication supplies the zero budget under RH. Neither direction replaces universal admissibility by evaluation of a Boolean program [7].

The separate finite-certificate objective asks for an explicitly defined type of finite raw data and a terminating checker satisfying

$$\big(\exists W : \mathrm{FiniteWitness},\ \mathrm{check}(W) = \mathrm{true}\big) \quad \Longleftrightarrow \quad \mathrm{RH}. \tag{68}$$

Soundness and completeness are logically distinct. A sufficient soundness theorem would extract an admissible cancellation budget from each acceptance. Composing this extraction with the semantic theorem would give RH. Completeness would still require a separate proof that RH implies the existence of accepted finite data. Neither checker theorem, nor an unconditional accepted witness, is supplied here.

Finite row computations can establish finitely many inequalities. To infer an infinite tail, a checker needs an additional proved mechanism, such as a certified recurrence with an invariant or a verified symbolic bound whose hypotheses its finite input establishes. This observation is a specification requirement, not an impossibility claim. A certificate may encode the finite reason for a universally valid argument, rather than list every instance of its conclusion. The precise arithmetic meaning of each encoded expression and the exact actual sequence must remain part of soundness. Merely placing a proof of the universal predicate inside a structure would provide mathematical evidence, but would not by itself define the requested raw-data format and Boolean acceptance procedure.

5.6 Conservative formalization targets

The next formalization tasks can be stated without enlarging the proved endpoint. First, starting from the probability measure already obtained under H_{00}, construct the complex L^2 space, the class of $\sqrt{w_a}$, and multiplication by λ_a. The target theorem is the concrete package

$$a > 0 \implies 0 \le T_a = T_a^*, \quad \|T_a\| \le 1, \quad \langle v_a, T_a^n v_a \rangle = \widetilde{\mu}_n(a). \tag{69}$$

Here operator positivity means positivity of every quadratic form. Measurability, square integrability, descent to almost-everywhere equivalence classes, and boundedness must be proved explicitly. The resolvent statement then has the conservative form

$$|h| < 1 \implies \langle v_a, (I - hT_a)^{-1} v_a \rangle = \int \frac{w_a}{1 - h\lambda_a}\, d\sigma. \tag{70}$$

Neither theorem mentions the actual Widder tower.

Second, connect this construction to the audited real Gram interface. One may use real-valued L^2 functions for the orbit argument, or take the real part of the complex inner product on the underlying real space. The required squared Cauchy inequality then follows from complex Cauchy–Schwarz. There is also a domain issue: the archived orbit interface quantifies over every real anchor, while these kernel inequalities assume $a > 0$. A positive-anchor reindexing, or an explicitly defined zero extension of the model family, must resolve that mismatch. Such an extension says nothing about actual coefficients at other anchors.

Third, if the longer Widder route is pursued, the additional target is

$$H_{00} \implies \forall a > 0\ \forall n, \quad \mu_n^{\mathrm{act}}(a) = \widetilde{\mu}_n(a). \tag{71}$$

A proof must relate the actual logarithmic derivative to the measure formula and justify the required differentiations and recursion at every order. Positivity of the model alone cannot replace those identities.

Finally, for a proposed finite checker, define its syntax and evaluation before seeking the two separate theorem targets

$$\mathrm{check}(W) = \mathrm{true} \Rightarrow \mathrm{RH}, \qquad \mathrm{RH} \Rightarrow \exists W,\ \mathrm{check}(W) = \mathrm{true}. \tag{72}$$

An accepted certificate would then have exactly the meaning proved by soundness. These are development targets, not additional audited results. The paper-level constructions on pages 39–41 do not modify the frozen Lean sources or their audit boundary at `ZETA_HYPERBOLIC_ANCHOR_5`. The distinction between a mathematical proof presented here and a newly checked Lean declaration is maintained throughout.

Appendix A.1 Verification evidence and its limits

The formal evidence refers to a fixed source baseline, not to every sentence in this expanded exposition. GitHub Actions run 40712534224 completed successfully at source commit

`c6b95fd9e3d792270026da083bddddbc72bacb22`.

The preserved record reports source-identity checks, the full project build, four exact endpoint type checks, and a strict audit of 641 selected declarations. The run's complete job log and final audit block were independently read when that checkpoint was recorded. The corresponding evidence files remain in the repository.

Property	Recorded value
Lean toolchain	`leanprover/lean4:v4.33.0`
Preserved Lean files	118 (117 modules and the final audit)
Audited declarations	641
Exact endpoint checks	Four, all passed
Active compiler target	`GITHUB_CI`
Local Lean execution for this manuscript	None

The mathlib revision is pinned to

`db584cd6d46c92f209a44c0f1c829460d327499d`.

The allowed axiom set is `propext`, `Classical.choice`, and `Quot.sound`. Each of the four principal endpoints has exactly this recorded set. The audit parser rejects missing or duplicate target entries and axioms outside the allowlist. This is evidence for the checked declarations and their dependencies, not a replacement for reading their types and understanding the meaning of their premises.

A separate native checkpoint records a user-reported Windows Lean 4.33.0 PASS. The original full native logs and native dependency revision were not independently supplied. The native and GitHub records are therefore retained separately; matching toolchain labels alone do not establish identical dependency revisions.

The present expansion changes manuscript files only. It does not modify the frozen Lean sources, discharge $H00$, or add a formal implementation of the displayed complex L^2 operator model. The latter is a conventional mathematical realization whose source formalization remains a distinct extension. Historical comments in the unchanged modules describe their preparation state before the later PASS records; the preserved evidence identifies the subsequent verification baseline precisely.

Appendix A.2 Exact endpoints and semantic assumptions

The four exported declarations lie in the namespace `RH_HilbertLimit.TwoRoutes`. With that namespace suppressed, their checked names and types are:

```
nativeRH_of_H00_route_one
  : H00 → _root_.RiemannHypothesis

H00_iff_nativeRH_route_one
  : H00 ↔ _root_.RiemannHypothesis

nativeRH_of_H00_route_two
  : H00 → _root_.RiemannHypothesis

H00_iff_nativeRH_route_two
  : H00 ↔ _root_.RiemannHypothesis
```

Here the conclusion is mathlib's literal global proposition, rather than a newly named project-local surrogate. The premise is the definition

```
H00 := zeroBudgetCandidate.Admissible.
```

Its expanded mathematical content is the zero-cancellation condition for every actual row. Since $C_n \geq 0$ unconditionally, the zero-budget inequality $C_n \leq 0$ is equivalent to $C_n = 0$ for every n. The direct finite identities then give

$$H00 \iff (\forall n,\ C_n = 0) \iff (\forall n,\ V_n = 1) \iff (\forall N, j \leq N,\ w_{N,j} \geq 0).$$

These equivalences identify the hypothesis; they do not establish it for the actual sequence. In particular, the data pair $(0, 0)$ being finite does not make its semantic admissibility a terminating check.

The two forward constructions have a shared beginning and distinct closures. Route 1 takes constant variation to subexponential growth and invokes the constructed actual spectral closure. Route 2 takes row positivity to a probability measure, obtains the actual logarithmic-derivative germ, and invokes analytic zero exclusion. Both reverse implications use the retained theorem `H00_of_nativeRH`. Thus two equivalence names do not assert two independent proofs of every supporting lemma.

For interpretation of other audited declarations, their hypotheses remain decisive. A theorem about a supplied `BKQRContractiveOrbit` proves consequences of that structure; it does not construct the structure for the actual Widder family. Similarly, a theorem whose premise is `ActualApproximationConvergence` does not remove that premise. The audit confirms proof terms for the stated propositions, while the manuscript's claim correspondence tracks which objects and assumptions those propositions describe.

Appendix B.1 What a finite positive prefix can establish

Finite rows provide necessary tests for an infinite moment sequence. They do not, for an unrestricted sequence, force positivity of all later rows. The following elementary example makes the quantifiers explicit.

Proposition 32. *For every integer $K \geq 0$, there is a real normalized sequence whose Bernstein rows of degrees $0, \ldots, K$ are nonnegative but whose row of degree $K+1$ has a negative entry.*

Proof. Set

$$u_0 = 1, \qquad u_1 = \cdots = u_K = 0, \qquad u_{K+1} = -1,$$

and, for definiteness, set all later terms to zero. When $K = 0$, the middle condition is empty. A row of degree N only uses the terms $u_0, \ldots, u_N$. Hence for $N \leq K$ its weights agree with those of the point mass δ_0:

$$w_{N,0} = 1, \qquad w_{N,j} = 0 \quad (1 \leq j \leq N).$$

Every such row is nonnegative and has variation one. In the next row, the last weight is

$$w_{K+1,K+1} = u_{K+1} = -1 < 0,$$

which proves the assertion. □

The example also has $C_n = 0$ for $n < K$, because those transitions join two rows of variation one. The next variation is strictly greater than one: row mass is still one but at least one weight is negative. The identity $C_K = (K+1)(V_{K+1} - V_K)$ therefore detects a strictly positive cancellation at the first unchecked transition.

This obstruction is about arbitrary extensions of finite scalar data. It does not prove that no finite certificate can ever establish a universal property of a fixed arithmetic sequence. A successful certificate could include additional structure, an induction invariant, or a theorem controlling every remaining row. What the example excludes is the inference from a positive prefix alone, without such a propagation argument.

For the actual ξ sequence, a negative exact weight would refute $H00$, whereas any finite list of nonnegative weights is consistent with both the desired universal assertion and failure farther out. Numerical approximations introduce a further distinction: a displayed small number is not an exact sign proof unless its error is controlled. The present endpoints start from semantic $H00$ and do not supply a finite algorithm deciding it. This is why the finite-checker program in Part 5 remains an additional theorem target.

Appendix B.2 Quantitative approximation of the resolvent

The moment construction also gives useful quantitative estimates once the representing measure is available. Define $\sigma_N = \sum_{j=0}^{N} w_{N,j}\delta_{j/N}$ for $N \geq 1$. Using the moment formula in each Bernstein weight gives

$$\int f\, d\sigma_N = \int_0^1 B_N f(q)\, d\sigma(q), \qquad B_N f(q) = \sum_{j=0}^{N} \binom{N}{j} q^j (1-q)^{N-j} f(j/N).$$

For fixed q, this is the expectation of $f(X/N)$ with X binomial of parameters N, q. If f is complex-valued and L-Lipschitz on $[0,1]$, Cauchy–Schwarz and the binomial variance give

$$|B_N f(q) - f(q)| \leq L\,\mathbb{E}|X/N - q| \leq L\sqrt{\frac{q(1-q)}{N}} \leq \frac{L}{2\sqrt{N}}.$$

Integrating proves the same upper bound for $|\int f\, d\sigma_N - \int f\, d\sigma|$. For continuous f, uniform approximation by Lipschitz functions yields weak convergence $\sigma_N \Rightarrow \sigma$. This supplies an ordinary limiting interpretation of the measure after its construction.

Proposition 33. *Let K be a nonempty compact subset of $\mathbb{C} \setminus (-\infty, -1/\kappa]$. Put*

$$d = \min_{z\in K,\, q\in[0,1]} |1 + \kappa q z| > 0, \qquad M = \max_{z\in K} |z|.$$

For $S_{\kappa,N}(z) = \kappa \int (1+\kappa q z)^{-1}\, d\sigma_N(q)$,

$$\sup_{z\in K} |S_{\kappa,N}(z) - S_\kappa(z)| \leq \frac{\kappa^2 M}{2d^2\sqrt{N}}.$$

Proof. For $f_z(q) = \kappa/(1+\kappa q z)$, the derivative with respect to the real variable q has modulus at most $\kappa^2 M/d^2$. The fundamental theorem of calculus makes this a uniform Lipschitz bound on the interval. Apply the preceding estimate for each z. □

Thus positive finite rows approximate the scalar resolvent uniformly on compact subsets of its common domain. The constants deteriorate near the slit because denominators can become small. This theorem does not identify the distinct weighted Widder coefficients: it concerns the measure resolvent built from the already specified moments. The estimate is a paper-level consequence of that measure representation, not an additional claim about the recorded Lean audit.

References

[1] J. B. Conway, *A Course in Functional Analysis*, second edition, Graduate Texts in Mathematics 96, Springer-Verlag, New York, 1990. doi:10.1007/978-1-4757-4383-8.

[2] F. Hausdorff, Summationsmethoden und Momentfolgen. I, *Mathematische Zeitschrift* **9** (1921), 74–109. doi:10.1007/BF01378337.

[3] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction—CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, pp. 625–635. doi:10.1007/978-3-030-79876-5_37.

[4] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, ACM, 2020, pp. 367–381. doi:10.1145/3372885.3373824.

[5] W. Rudin, *Real and Complex Analysis*, third edition, McGraw-Hill, New York, 1987. ISBN 978-0-07-054234-1.

[6] E. C. Titchmarsh, *The Theory of the Riemann Zeta-function*, second edition, revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986. ISBN 0-19-853369-1.

[7] *Hilber-Vector-Resolvent-Theory*, project source repository, frozen construction package `RH_H0O_TWO_ROUTES_LEAN_v0_2_3_construction`. Audited source commit `c6b95fd9e3d792270026da083bddddbc72bacb22`. Repository and manuscript sources; GitHub CI run 40712534224. Verification records dated 28 October 2026.

The classical references provide background for the Hilbert-space, moment, and analytic constructions. The source reference identifies the exact project statements and their proof dependencies. The Lean and mathlib publications describe the proof infrastructure; the specific dependency versions used by this project are the pins in Appendix A. No new bibliography entry is intended to replace a source-level audit of the actual conditional endpoints.

Appendix C. Source concordance and reproducibility

The table locates the main mathematical stages. Paths in the middle column are relative to `RH_HilbertLimit/TwoRoutes/`, except where an explicit parent path is shown. The source baseline is the commit recorded in Appendix A.

Manuscript topic	Source module or record	Role
Hilbert defects	`Derivations/BKQR/DefectGramSquare.lean`	Abstract real Gram interface
$H0$0 and row positivity	`RouteOne.lean`; `ActualPositivePrefix.lean`	Actual conditional identities
Monomial limit	`BernsteinMomentLimit.lean`	General real sequences
Positive measure	`HausdorffMeasure.lean`	Constructed from positivity
Kernel holomorphy	`ResolventAnalytic.lean`	Local denominator control
Actual germ	`MeasureResolvent.lean`	Exact derivative identification
Continuation	`ActualResolventContinuation.lean`	Analytic defect argument
Pole closure	`ActualHerglotzClosure.lean`	Sign implies actual RH
Weighted frontier	`RouteTwoContractive.lean`; `ActualLimitBridge.lean`	Bounds and explicit open bridge
Endpoint types	`ci/EndpointTypes.lean`	Four exact type checks
Audit evidence	`evidence/GITHUB_CI_AXIOM_AUDIT.json`	641 declaration results

The spectral route also uses the retained `RH_HilbertLimit/` modules. These include `AnalyticClosure.lean` and the actual calibration and extraction infrastructure. The finer introduction source map under `Paper-Candidates/` records the difference between the minimal real Gram interface and the concrete complex Hilbert-space realization used in this exposition.

To rebuild this manuscript, run `python build.py` in its source directory with Python, `pdflatex`, and Poppler's `pdfinfo` available. The builder invokes only LaTeX, checks the expected subsection-to-page map, rejects overflow and unresolved references, and verifies the fifty-page output. Lean compilation remains assigned to the existing GitHub workflow. Rebuilding the PDF is not a new formal-verification checkpoint.

The next mathematical extensions have separate acceptance criteria: construct the concrete operator model in Lean; prove the exact Widder identification if the longer route is pursued; and establish a universal admissibility argument before asserting an accepted certificate or unconditional RH. Each would change the theorem coverage and should carry its own source and verification record.